

Ministerul Educației și Cercetării al Republicii Moldova
Universitatea Tehnică a Moldovei
Facultatea Calculatoare, Informatică și Microelectronică

Raport

la

Lucrarea de laborator Nr. 1

Disciplina: Tehnici de programare

Tema: ”Linux Luminarium”

A efectuat :

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A verificat :

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Chișinău – 2026

SARCINA LUCRĂRII

1. Hello Hackers:

1. Intro to Commands:

```
hacker@hello~intro-to-commands:~$ hello
Success! Here is your flag:
pwn.college{MaqbQaMaR4_wKfVHFdr_ostVFyg.QX3YjM1wyN5QTMxIzW}
hacker@hello~intro-to-commands:~$
```

🎉 You've already solved Intro to Commands! 🎉

>_ Terminal [↗](#) </> Code [↗](#) 🖥 Desktop [↗](#) 🔑 SSH [ⓘ](#)

🚩 /Fyg.QX3YjM1wyN5QTMxIzW} 🔁 ☐ sudo

2. Intro to Arguments:

```
hacker@hello~intro-to-arguments:~$ hello hackers
Success! Here is your flag:
pwn.college{4eUophVZJ_wARubfoMU5J2o_hcB.QX4YjM1wyN5QTMxIzW}
hacker@hello~intro-to-arguments:~$
```

🎉 You've already solved Intro to Arguments! 🎉

>_ Terminal

</> Code

Desktop

SSH

_hcB.QX4YjM1wyN5QTMxIzW}

↺

sudo

3. Command History:

```
hacker@hello~command-history:~$ the flag is pwn.college{QjRihHfF7rmyF6S8XJQYKtSZ  
t64.0lNzEzNxwyN5QTMxIzW}
```

🎉 You've already solved Command History! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 t64.0lNzEzNxwyN5QTMxIzW}

↻

📄 sudo

2. Pondering Paths:

1. The Root:

```
hacker@paths~the-root:~$ /pwn
BOOM!!!
Here is your flag:
pwn.college{EH3lSj6vN0BvejQfXWcRtdraBSI.QX4cT00wyN5QTMxIzW}
hacker@paths~the-root:~$
```

🎉 You've already solved The Root! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 aBSI.QX4cT00wyN5QTMxIzW}



sudo

2. Program and absolute paths:

```
hacker@paths~program-and-absolute-paths:~$ /challenge/run
Correct!!!
/challenge/run is an absolute path! Here is your flag:
pwn.college{cy-25vWwQ9VrQHY5Au_IfrdpFdD.QX1QTN0wyN5QTMxIzW}
hacker@paths~program-and-absolute-paths:~$
```

🎉 You've already solved Program and absolute paths! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 IfrdpFdD.QX1QTN0wyN5QTMxIzW}

↻

📄 sudo

3. Position thy self:

```
hacker@paths~position-thy-self:~$ /challenge/run
Incorrect...
You are not currently in the /usr/include directory.
Please use the `cd` utility to change directory appropriately.
hacker@paths~position-thy-self:~$ cd /usr/include
hacker@paths~position-thy-self:/usr/include$ /challenge/run
Correct!!!
/challenge/run is an absolute path, invoked from the right directory!
Here is your flag:
pwn.college{UCbsF1IJkI9MLLYuUI1-HikUkn_.QX2QTN0wyN5QTMxIzW}
hacker@paths~position-thy-self:/usr/include$
```

🎉 You've already solved Position thy self! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 kn_.QX2QTN0wyN5QTMxIzW}

↻

📄 sudo

4. Position elsewhere:

```
hacker@paths~position-elsewhere:~$ /challenge/run
Starting level 1.
Incorrect...
You are not currently in the /etc directory.
Please use the `cd` utility to change directory appropriately.
hacker@paths~position-elsewhere:~$ cd /etc
hacker@paths~position-elsewhere:/etc$ /challenge/run
Starting level 1.
Correct!!!
/challenge/run is an absolute path, invoked from the right directory!
Moving on to level 2
Please use the `cd` utility to change directory to /var
hacker@paths~position-elsewhere:/etc$ cd /var
hacker@paths~position-elsewhere:/var$ /challenge/run
Correct!!!
/challenge/run is an absolute path, invoked from the right directory!
Moving on to level 3
Please use the `cd` utility to change directory to /sys
hacker@paths~position-elsewhere:/var$ cd /sys
hacker@paths~position-elsewhere:/sys$ /challenge/run
Correct!!!
/challenge/run is an absolute path, invoked from the right directory!
Moving on to level 4
Please use the `cd` utility to change directory to /usr/bin
hacker@paths~position-elsewhere:/sys$ cd /usr/bin
hacker@paths~position-elsewhere:/usr/bin$ /challenge/run
Correct!!!
/challenge/run is an absolute path, invoked from the right directory!
Moving on to level 5
Please use the `cd` utility to change directory to /var/lib/apt/lists
hacker@paths~position-elsewhere:/usr/bin$ /var/lib/apt/lists
bash: /var/lib/apt/lists: Is a directory
hacker@paths~position-elsewhere:/usr/bin$ cd /var/lib/apt/lists
hacker@paths~position-elsewhere:/var/lib/apt/lists$ /challenge/run
Correct!!!
/challenge/run is an absolute path, invoked from the right directory!
Here is your flag:
pwn.college{4fpZwfb5Ex_ytv42pG-2BiNibWW.QX3QTN0wyN5QTMxIzW}
hacker@paths~position-elsewhere:/var/lib/apt/lists$
```

 You've already solved Position elsewhere! 

> Terminal

</> Code

Desktop

SSH

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↺

□ sudo

5. implicit relative paths, from /:

```
hacker@paths~implicit-relative-paths-from-:~$ cd /  
hacker@paths~implicit-relative-paths-from-:/$ challenge/run  
Correct!!!  
challenge/run is a relative path, invoked from the right directory!  
Here is your flag:  
pwn.college{AwZER0A-lw8BpyA6V8_yrR9LfRo.QX5QTN0wyN5QTMxIzW}  
hacker@paths~implicit-relative-paths-from-:/$
```

🎉 You've already solved implicit relative paths,
from /! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 .fRo.QX5QTN0wyN5QTMxIzW}



📄 sudo

6. explicit relative paths, from:

```
hacker@paths~explicit-relative-paths-from-:~$ cd /  
hacker@paths~explicit-relative-paths-from-:/$ ./challenge/run  
Correct!!!  
./challenge/run is a relative path, invoked from the right directory!  
Here is your flag:  
pwn.college{cpfqtnDV0ptZxTZupSicavDkjVX.QXwUTN0wyN5QTMxIzW}  
hacker@paths~explicit-relative-paths-from-:/$
```

🎉 You've already solved explicit relative paths,
from /! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 jVX.QXwUTN0wyN5QTMxIzW}



📄 sudo

7. implicit relative path:

```
hacker@paths-implicit-relative-path:~$ cd /challenge
hacker@paths-implicit-relative-path:/challenge$ ./run
Correct!!!
./run is a relative path, invoked from the right directory!
Here is your flag:
pwn.college{0N4_9D51Cu9uiPii4JcvAH20TkI.QXxUTN0wyN5QTMxIzW}
hacker@paths-implicit-relative-path:/challenge$
```

🎉 You've already solved implicit relative path! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 0TkI.QXxUTN0wyN5QTMxIzW}

↻

📄 sudo

8. home sweet home:

```
hacker@paths-home-sweet-home:~$ /challenge/run ~/a
Writing the file to /home/hacker/a!
... and reading it back to you:
pwn.college{YHsXifepgnjZBaNZSfpIn0w-uP9.QXzMD00wyN5QTMxIzW}
hacker@paths-home-sweet-home:~$
```

🎉 You've already solved home sweet home! 🎉

> Terminal

</> Code

Desktop

SSH

🚩 p9.QXzMD00wyN5QTMxIzW}



sudo

3. Comprehending Commands:

1. cat: not the pet, but the command!:

```
hacker@commands-cat-not-the-pet-but-the-command:~$ cat flag  
pwn.college{cg8TQSHvonDogfini326p0syZNP.QXxcTN0wyN5QTMxIzW}  
hacker@commands-cat-not-the-pet-but-the-command:~$
```

🎉 You've already solved cat: not the pet, but the command!! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 |ZNP.QXxcTN0wyN5QTMxIzW}

↻

📄 sudo

2. catting absolute paths:

```
hacker@commands~catting-absolute-paths:~$ cat /flag
pwn.college{ILNg88YEZvNTamlu6m0yL3s2J0l.QX5ET00wyN5QTMxIzW}
hacker@commands~catting-absolute-paths:~$
```

🎉 You've already solved catting absolute paths! 🎉

>_ Terminal

</> Code

Desktop

SSH

2J0l.QX5ET00wyN5QTMxIzW}

↻

□ sudo

3. more catting practice:

You cannot use the 'cd' command in this level, and must retrieve the flag by absolute path. Plus, I hid the flag in a different directory! You can find it in the file /usr/share/util-linux/flag. Go cat it out without using cd!

```
hacker@commands~more-catting-practice:~$ cat /usr/share/util-linux/flag
```

```
pwn.college{8Ua3MUy6tYph0eHVSRYwDNhd0th.QXwIT00wyN5QTMxIzW}
```

```
hacker@commands~more-catting-practice:~$
```

🎉 You've already solved more catting practice! 🎉

>_ Terminal

</> Code

Desktop

SSH

🚩 d0th.QXwIT00wyN5QTMxIzW}



sudo

4. grepping for a needle in a haystack:

```
hacker@commands~grepping-for-a-needle-in-a-haystack:~$ grep pwn.college /challenge/data.txt
pwn.college{wwrQaoUz9YPh_CiC8tcmRAQdx43.QX3ED00wyN5QTMxIzW}
hacker@commands~grepping-for-a-needle-in-a-haystack:~$
```

🎉 You've already solved grepping for a needle in a haystack! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 x43.QX3ED00wyN5QTMxIzW}



📄 sudo

5. comparing files:

```
hacker@commands~comparing-files:~$ diff /challenge/decoys_only.txt /challenge/decoys_and_real.txt
20a21
> pwn.college{IGt4u2ruD1FaAGnGhQ0uIK_HNNM.01MwMDOxwyN5QTMxIzW}
```

🎉 You've already solved comparing files! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 V.01MwMDOxwyN5QTMxIzW}

↻

📄 sudo

6. listing files:

```
hacker@commands~listing-files:~$ ls /challenge
5558-renamed-run-15651 Dockerfile
hacker@commands~listing-files:~$ /challenge/5558-renamed-run-15651
Yahaha, you found me! Here is your flag:
pwn.college{k_quzJF65qRBfIm2xndZGNxIgad.QX4ID00wyN5QTMxIzW}
hacker@commands~listing-files:~$
```

🎉 You've already solved listing files! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 Igad.QX4ID00wyN5QTMxIzW}



📄 sudo

7. touching files:

```
hacker@commands~touching-files:~$ touch /tmp/pwn
hacker@commands~touching-files:~$ touch /tmp/college
hacker@commands~touching-files:~$ /challenge/run
Success! Here is your flag:
pwn.college{0y2y6zUdOCBUKoEad4xRIrJ3YCN.QXwMD00wyN5QTMxIzW}
hacker@commands~touching-files:~$
```

🎉 You've already solved touching files! 🎉

>_ Terminal

</> Code

Desktop

SSH

🚩 CN.QXwMD00wyN5QTMxIzW}

↺

📄 sudo

8. removing files:

```
hacker@commands-removing-files:~$ rm delete_me
hacker@commands-removing-files:~$ /challenge/check
Excellent removal. Here is your reward:
pwn.college{0627fViNYvEFkSyo0zkS4k6d1HJ.QX2kDM1wyN5QTMxIzW}
hacker@commands-removing-files:~$
```

🎉 You've already solved removing files! 🎉

>_ Terminal

</> Code

Desktop

SSH

1HJ.QX2kDM1wyN5QTMxIzW}



sudo

9. moving files:

```
hacker@commands-moving-files:~$ mv /flag /tmp/hack-the-planet
Correct! Performing 'mv /flag /tmp/hack-the-planet'.
hacker@commands-moving-files:~$ /challenge/check
Congrats! You successfully moved the flag to /tmp/hack-the-planet! Here it is:
pwn.college{A8y4cAX0kbxpuSxcJtXMmY7n-r6.0V0xEzNxwyN5QTMxIzW}

hacker@commands-moving-files:~$
```

🎉 You've already solved moving files! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 -r6.0V0xEzNxwyN5QTMxIzW}



📄 sudo

10. copying files:

```
hacker@commands-copying-files:~$ cp /flag /tmp/hack-the-planet
Correct! Performing 'cp /flag /tmp/hack-the-planet'.
hacker@commands-copying-files:~$ /challenge/check
Congrats! You successfully copied the flag to /tmp/hack-the-planet! Here it is:
pwn.college{k3v51oYdKKEaI8JZ2ahh17eX20E.0lNxQTMwyN5QTMxIzW}

hacker@commands-copying-files:~$
```

🎉 You've already solved copying files! 🎉

>_ Terminal

</> Code

Desktop

SSH

!OE.OlNxQTMwyN5QTMxIzW}



sudo

11. hidden files:

```
hacker@commands-hidden-files:~$ ls -a /  
.  
..  
.dockerenv  
.flag-2223017837503  
bin  
boot  
challenge  
dev  
etc  
home  
lib  
lib64  
media  
mnt  
nix  
opt  
proc  
root  
run  
sbin  
srv  
sys  
tmp  
usr  
var  
hacker@commands-hidden-files:~$ cat /.flag-2223017837503  
pwn.college{4he4Hz0YL1QN_N0i842MuX-pPa-.QXwUD00wyN5QTMxIzW}  
hacker@commands-hidden-files:~$
```

🎉 You've already solved hidden files! 🎉

> Terminal

</> Code

Desktop

SSH

Pa-.QXwUD00wyN5QTMxIzW}

↻

sudo

12. An Epic Filesystem Quest:

```
hacker@commands-an-epic-filesystem-quest:~$ cd /
hacker@commands-an-epic-filesystem-quest:/$ ls
REVELATION  boot      dev  flag  lib    media  nix  proc  run  srv  tmp  var
bin         challenge etc  home  lib64  mnt    opt  root  sbin sys  usr
hacker@commands-an-epic-filesystem-quest:/$ cat REVELATION
Lucky listing!
The next clue is in: /usr/lib/python3/dist-packages/pip/_internal/commands

Watch out! The next clue is **trapped**. You'll need to read it out without 'cd'ing into
the directory; otherwise, the clue will self destruct!
hacker@commands-an-epic-filesystem-quest:/$ ls /usr/lib/python3/dist-packages/pip/_intern
al/commands
WHISPER-TRAPPED  check.py      download.py  index.py     search.py
__init__.py     completion.py freeze.py     inspect.py   show.py
__pycache__     configuration.py hash.py       install.py   uninstall.py
cache.py        debug.py      help.py      list.py      wheel.py
hacker@commands-an-epic-filesystem-quest:/$ cat /usr/lib/python3/dist-packages/p
ip/_internal/commands/WHISPER-TRAPPED
Lucky listing!
The next clue is in: /usr/share/doc/zlib1g

The next clue is **hidden** --- its filename starts with a '.' character. You'll need to
look for it using special options to 'ls'.
hacker@commands-an-epic-filesystem-quest:/$ ls -a /usr/share/doc/zlib1g
.  .. .MESSAGE changelog.Debian.gz  copyright
hacker@commands-an-epic-filesystem-quest:/$ cat /usr/share/doc/zlib1g/.MESSAGE
Congratulations, you found the clue!
The next clue is in: /usr/lib/python3.12/dbm

Watch out! The next clue is **trapped**. You'll need to read it out without 'cd'ing into
the directory; otherwise, the clue will self destruct!
hacker@commands-an-epic-filesystem-quest:/$ ls /usr/lib/python3.12/dbm
CLUE-TRAPPED  __init__.py  __pycache__  dumb.py  gnu.py  ndbm.py
hacker@commands-an-epic-filesystem-quest:/$ cat /usr/lib/python3.12/dbm/CLUE-TRAPPED
Lucky listing!
The next clue is in: /usr/share/doc/libc6

The next clue is **hidden** --- its filename starts with a '.' character. You'll need to
look for it using special options to 'ls'.
hacker@commands-an-epic-filesystem-quest:/$ ls -a /usr/share/doc/libc6
.  .INSIGHT  NEWS.gz  README.hesiod.gz  copyright
```

```

hacker@commands-an-epic-filesystem-quest:/$ cat /usr/share/doc/libc6/.INSIGHT
Tubular find!
The next clue is in: /usr/lib/x86_64-linux-gnu/perl/5.38.2/asm-generic

Watch out! The next clue is **trapped**. You'll need to read it out without 'cd'ing into
the directory; otherwise, the clue will self destruct!
hacker@commands-an-epic-filesystem-quest:/$ ls /usr/lib/x86_64-linux-gnu/perl/5.38.2/asm-
generic
CUE-TRAPPED      ioctl.ph  posix_types.ph  sockios.ph      termbits.ph
bitsperlong.ph  ioctl.ph  socket.ph       termbits-common.ph  termios.ph
hacker@commands-an-epic-filesystem-quest:/$ cat /usr/lib/x86_64-linux-gnu/perl/5.38.2/asm
-generic/CUE-TRAPPED
Great sleuthing!
The next clue is in: /usr/share/doc/libss2

The next clue is **delayed** --- it will not become readable until you enter the director
y with 'cd'.
hacker@commands-an-epic-filesystem-quest:/$ cd /usr/share/doc/libss2
hacker@commands-an-epic-filesystem-quest:/usr/share/doc/libss2$ ls
TRACE  changelog.Debian.gz  copyright
hacker@commands-an-epic-filesystem-quest:/usr/share/doc/libss2$ cat TRACE
Congratulations, you found the clue!
The next clue is in: /usr/include/linux/isdn

The next clue is **hidden** --- its filename starts with a '.' character. You'll need to
look for it using special options to 'ls'.
hacker@commands-an-epic-filesystem-quest:/usr/share/doc/libss2$ ls -a /usr/include/linux/
isdn
.  ..  .BRIEF  capicmd.h
hacker@commands-an-epic-filesystem-quest:/usr/share/doc/libss2$ cat /usr/include/linux/is
dn/.BRIEF
Tubular find!
The next clue is in: /usr/lib/x86_64-linux-gnu/perl-base/unicore/lib/Math
hacker@commands-an-epic-filesystem-quest:/usr/share/doc/libss2$ ls /usr/lib/x86_64-linux-
gnu/perl-base/unicore/lib/Math
README  Y.pl
hacker@commands-an-epic-filesystem-quest:/usr/share/doc/libss2$ cat /usr/lib/x86_64-linux
-gnu/perl
CONGRATUL
It is: p
hacker@cc

```



You've already solved An Epic Filesystem Quest!



>_ Terminal

</> Code

Desktop

SSH



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sudo

13. making directories:

```
hacker@commands~making-directories:~$ mkdir /tmp/pwn
hacker@commands~making-directories:~$ cd /tmp/pwn
hacker@commands~making-directories:/tmp/pwn$ touch college
hacker@commands~making-directories:/tmp/pwn$ cd
hacker@commands~making-directories:~$ /challenge/run
Success! Here is your flag:
pwn.college{UdqCeR9VtmGswb4SQhec4cXTbqD.QXxMD00wyN5QTMxIzW}
hacker@commands~making-directories:~$
```

🎉 You've already solved making directories! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 ›qD.QXxMD00wyN5QTMxIzW}

↻

📄 sudo

14. finding files:

```
hacker@commands-finding-files:~$ find / -name flag
find: '/root': Permission denied
find: '/etc/ssl/private': Permission denied
/usr/lib/python3/dist-packages/pwnlib/flag
/usr/lib/python3/dist-packages/nacl/bindings/__pycache__/flag
find: '/var/cache/apt/archives/partial': Permission denied
find: '/var/cache/ldconfig': Permission denied
find: '/var/lib/apt/lists/partial': Permission denied
find: '/proc/tty/driver': Permission denied
find: '/proc/1/task/1/fd': Permission denied
find: '/proc/1/task/1/fdinfo': Permission denied
find: '/proc/1/task/1/ns': Permission denied
find: '/proc/1/fd': Permission denied
find: '/proc/1/map_files': Permission denied
find: '/proc/1/fdinfo': Permission denied
find: '/proc/1/ns': Permission denied
find: '/proc/7/task/7/fd': Permission denied
find: '/proc/7/task/7/fdinfo': Permission denied
find: '/proc/7/task/7/ns': Permission denied
find: '/proc/7/fd': Permission denied
find: '/proc/7/map_files': Permission denied
find: '/proc/7/fdinfo': Permission denied
find: '/proc/7/ns': Permission denied
/nix/store/7ns27apnvn4qj4q5c82x0z1lzixrz47p-radare2-5.9.8/share/radare2/5.9.8/flag
/nix/store/5z3sjp9r463i3siif58hq5wj5jmy5m98-python3.12-pwntools-4.13.1/lib/python3.12/site-packages/pwnlib/flag
/nix/store/5n5lp1m8gilgrsriv1f2z0jdk50ypcn-rizin-0.7.3/share/rizin/flag
/nix/store/h88mxp2mbgyj06vypwmqpy05idhwimnp-python3.13-pwntools-4.14.1/lib/python3.13/site-packages/pwnlib/flag
/nix/store/s8b49lb0pqww0c6kgjbxdcv2bp0x4-radare2-5.9.8/share/radare2/5.9.8/flag
/nix/store/bnlabj2vsbljhp597ir29l51nrqhm89w-rizin-0.7.4/share/rizin/flag
/nix/store/1hyxipwpcxw90l5pq1nvd6s6jdi5m-python3.12-pwntools-4.14.1/lib/python3.12/site-packages/pwnlib/flag
/nix/store/5qz6hgb1qzpvjrs20wyiylx5zw8b9bk-pwntools-4.14.0/lib/python3.13/site-packages/pwnlib/flag
/nix/store/c07wba3nrck81kd0yg5h8rx22di35xi-radare2-6.0.4/share/radare2/6.0.4/flag
/nix/store/gfyhjlxav9nczmnab3jxr73kb22yp42-rizin-0.8.1/share/rizin/flag
/nix/store/61dd247b72i7xnm0mw9qaxgxf3gs2lyy-python3.13-pwntools-4.14.1/lib/python3.13/site-packages/pwnlib/flag
/nix/store/sh1s72wgvqc50kp830nlhm5cjpmyh7-python3.13-pwntools-4.14.1/lib/python3.13/site-packages/pwnlib/flag
/nix/store/1wn496frzskd2drjwyyskk3g50rd1nbd-pwntools-4.14.1/lib/python3.13/site-packages/pwnlib/flag
/nix/store/rszkfmr2mp5r4dyvhwcqj3521nyqfyzc-rizin-0.8.2/share/rizin/flag
/nix/store/p9vjrnqyl7hy1k0ik9dw0lb9ib3c4582-python3.13-pwntools-4.15.0/lib/python3.13/site-packages/pwnlib/flag
/nix/store/kdlmh9h9hm225jarh6rk927lb7jsxsv-radare2-6.1.8/share/radare2/6.1.8/flag
/nix/store/hss94rqc4ha0y229f8fwcy304wnzdsq7-pwntools-4.14.1/lib/python3.13/site-packages/pwnlib/flag
/nix/store/f42x3k5mrk3qq79ixwy991p6g6a5l9f6-python3.13-pwntools-4.15.0/lib/python3.13/site-packages/pwnlib/flag
/nix/store/4ckl0p6mafmm242c9qv3sl5dbd9rr9j-source/packages/core/src/flag
find: '/nix/var/nix/builds/nix-8-2086155677': Permission denied
find: '/nix/var/nix/builds/nix-8-4078488522': Permission denied
hacker@commands-finding-files:~$ cat /usr/lib/python3/dist-packages/nacl/bindings/__pycache__/_flag
pwn.college{swu8FtqceW0DN0CIk0sH2gDnyc6.QXyMD00wyN5QTMxIzW}hacker@commands-finding-files:
hacker@commands-finding-files:~$
```

Terminal

Code

Desktop

SSH



jc6.QXyMD00wyN5QTMxIzW}



sudo

15. linking files:

```
hacker@commands~linking-files:~$ ln -s /flag /home/hacker/not-the-flag
ln: failed to create symbolic link '/home/hacker/not-the-flag': File exists
hacker@commands~linking-files:~$ rm /home/hacker/not-the-flag
hacker@commands~linking-files:~$ ln -s /flag /home/hacker/not-the-flag
hacker@commands~linking-files:~$ /challenge/catflag
About to read out the /home/hacker/not-the-flag file!
pwn.college{wF1XwgMR_EHAfqtVHx2MIRUtXLZ.QX5ETN1wyN5QTMxIzW}
hacker@commands~linking-files:~$
```

🎉 You've already solved linking files! 🎉

> _ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 tXLZ.QX5ETN1wyN5QTMxIzW}



📄 sudo

4. Digesting Documentation:

1. Learning From Documentation:

```
hacker@man~learning-from-documentation:~$ /challenge/challenge --giveflag
Correct argument! Here is your flag:
pwn.college{gMZlhjElvoy6LCBUUORqfD8c5zY.QX0IT00wyN5QTMxIzW}
hacker@man~learning-from-documentation:~$
```

 You've already solved Learning From Documentation! 

 Terminal 

 Code 

 Desktop 

 SSH 

 c5zY.QX0IT00wyN5QTMxIzW}



 sudo

2. Learning Complex Usage:

```
hacker@man~learning-complex-usage:~$ /challenge/challenge --printfile /flag
Correct argument! Here is the /flag file:
pwn.college{opA5_HQI2Lt2H_OyH3bsHVsyA1K.QX1IT00wyN5QTMxIzW}
hacker@man~learning-complex-usage:~$
```

🎉 You've already solved Learning Complex Usage!



> Terminal

</> Code

Desktop

SSH

Ya1K.QX1IT00wyN5QTMxIzW}



sudo

3. Reading Manuals:

```
hacker@man~reading-manuals:~$ man challenge

CHALLENGE(1)                  Challenge Commands                  CHAL
LENGE(1)

NAME
    /challenge/challenge - print the flag!

SYNOPSIS
    challenge OPTION

DESCRIPTION
    Output the flag when called with the right arguments.

    --fortune
        read a fortune

    --version
        output version information and exit

    --gsboay NUM
        print the flag if NUM is 815

AUTHOR
    Written by Zardus.

REPORTING BUGS
    The repository for this dojo: <https://github.com/pwncollege/linux-lu-
    minarium/>

SEE ALSO
    man(1) bash-builtins(7)

pwn.college                    May 2024                    CHALLENGE
(1)
hacker@man~reading-manuals:~$ /challenge/challenge --gsboay 815
Correct usage! Your flag: pwn.college{gsb8oGa15Ly6BcrLGwa7vk0hUbI.QX0ED00wyN5QTM
xIzW}
```

🎉 You've already solved Reading Manuals! 🎉

> Terminal

</> Code

Desktop

SSH

🚩 iUbI.QX0ED00wyN5QTMxIzW}

↻

📄 sudo

4. Searching Manuals:

```
--kk This argument will give you the flag!
--huu A neat argument, but not the right one.
--baicf A neat argument, but not the right one.
--sn A neat argument, but not the right one.
--rtj A neat argument, but not the right one.
--drtmdf A neat argument, but not the right one.
--nvupwh A neat argument, but not the right one.
--bkqbj A neat argument, but not the right one.
--mec A neat argument, but not the right one.
--scmapar A neat argument, but not the right one.
--fh A neat argument, but not the right one.
--auuwtwj A neat argument, but not the right one.
--vipxb A neat argument, but not the right one.
--iku A neat argument, but not the right one.
--nxgxxaf A neat argument, but not the right one.
--dhgke A neat argument, but not the right one.
```

Manual page challenge(1) line 1809/2035 91% (press h for help or q

```
hacker@man-searching-manuals:~$ man challenge
```

```
gzip: stdout: Broken pipe
```

```
hacker@man-searching-manuals:~$ /challenge/challenge --k
```

```
Initializing...
```

```
Incorrect usage! Please read the challenge man page!
```

```
hacker@man-searching-manuals:~$ /challenge/challenge --kk
```

```
Initializing...
```

```
Correct usage! Your flag: pwn.college{gngf1bP41l7T2T2xsu5KgIJUzkh.QX1ED00wyN5QTMxIzW}
```

```
hacker@man-searching-manuals:~$
```

🎉 You've already solved Searching Manuals! 🎉

> Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 Jzkh.QX1ED00wyN5QTMxIzW}

↻

📄 sudo

5. Searching for Manuals:

```
hacker@man-searching-for-manuals:~$ man -k challenge
tnxcupkkfw (1)      - print the flag!
hacker@man-searching-for-manuals:~$ man tnxcupkkfw

CHALLENGE(1)          Challenge Commands          CHAL
LENGE(1)

NAME
    /challenge/challenge - print the flag!

SYNOPSIS
    challenge OPTION

DESCRIPTION
    Output the flag when called with the right arguments.

    --fortune
        read a fortune

    --version
        output version information and exit

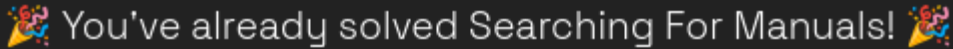
    --tnxcup NUM
        print the flag if NUM is 471

AUTHOR
    Written by Zardus.

REPORTING BUGS
    The repository for this dojo: <https://github.com/pwncollege/linux-luminarium/>

SEE ALSO
    man(1) bash-builtins(7)

pwn.college          May 2024          CHALLENGE
(1)
hacker@man-searching-for-manuals:~$ /challenge/challenge --tnxcup 471
Correct usage: Your flag must follow 64bGvDwG7Ww5TE3LUNGGDQ02_QX2ED00wYn5QTMxIzW}
hacker@ma
```



 Terminal 

 Code 

 Desktop 

 SSH 

 OC2.QX2ED00wyN5QTMxIzW}



 sudo

6. Helpful Programs:

```
hacker@man-helpful-programs:~$ /challenge/challenge --help
usage: a challenge to make you ask for help [-h] [--fortune] [-v]
      [-g GIVE_THE_FLAG] [-p]

options:
  -h, --help            show this help message and exit
  --fortune             read your fortune
  -v, --version         get the version number
  -g GIVE_THE_FLAG, --give-the-flag GIVE_THE_FLAG
                        get the flag, if given the correct value
  -p, --print-value     print the value that will cause the -g option to give
                        you the flag

hacker@man-helpful-programs:~$ /challenge/challenge -g GIVE_THE_FLAG
usage: a challenge to make you ask for help [-h] [--fortune] [-v]
      [-g GIVE_THE_FLAG] [-p]
a challenge to make you ask for help: error: argument -g/--give-the-flag: invalid
int value: 'GIVE_THE_FLAG'
hacker@man-helpful-programs:~$ /challenge/challenge -p
The secret value is: 888
hacker@man-helpful-programs:~$ /challenge/challenge -g 888
Correct usage! Your flag: pwn.college{kz8AL_e8gxzNINaK8fDqaCRKSLp.QX3ID00wyN5QTM
xIzW}
hacker@man-helpful-programs:~$
```

 You've already solved Helpful Programs! 

 Terminal 

 Code 

 Desktop 

 SSH 

 {SLp.QX3ID00wyN5QTMxIzW}



 sudo

7. Help for Builtins:

```
hacker@man-help-for-builtins:~$ help challenge
challenge: challenge [--fortune] [--version] [--secret SECRET]
  This builtin command will read you the flag, given the right arguments!

Options:
  --fortune      display a fortune
  --version      display the version
  --secret VALUE prints the flag, if VALUE is correct

You must be sure to provide the right value to --secret. That value
is "g87RI69s".
hacker@man-help-for-builtins:~$ challenge --secret g87RI69s
Correct! Here is your flag!
pwn.college{g87RI69sgjxIioZ60BmkiZsSoqb.QX0ET00wyN5QTMxIzW}

hacker@man-help-for-builtins:~$
```

🎉 You've already solved Help for Builtins! 🎉

>_ Terminal

</> Code

Desktop

SSH

oqb.QX0ET00wyN5QTMxIzW}



sudo

5. File Globbing:

1. Matching with *.png:

```
This challenge resets your working directory to /home/hacker unless you change
directory properly...
This challenge resets your working directory to /home/hacker unless you change
directory properly...
hacker@globbing-matching-with-:~$ cd /ch*
hacker@globbing-matching-with-:/challenge$ /challenge/run
You ran me with the working directory of /challenge! Here is your flag:
pwn.college{AdgbEJ_pdI4dNwqsE00Lw5w5Qpy.QXxID00wyN5QTMxIzW}
hacker@globbing-matching-with-:/challenge$
```

🎉 You've already solved Matching with *! 🎉

>_ Terminal

</> Code

Desktop

SSH

🚩 iQpy.QXxID00wyN5QTMxIzW}

↻

📄 sudo

2. Matching with ?:

```
This challenge resets your working directory to /home/hacker unless you change
directory properly...
This challenge resets your working directory to /home/hacker unless you change
directory properly...
hacker@globbing~matching-with-:~$ cd /?ha??enge
hacker@globbing~matching-with-:/challenge$ /challenge/run
You ran me with the working directory of /challenge! Here is your flag:
pwn.college{Q40VQYApi54d6jnMPiS-P-PBQDk.QXyID00wyN5QTMxIzW}
hacker@globbing~matching-with-:/challenge$
```

🎉 You've already solved Matching with ?! 🎉

>_ Terminal

</> Code

Desktop

SSH

🚩 }QDk.QXyID00wyN5QTMxIzW}

↻

📄 sudo

3. Matching with []:

```
hacker@globbing~matching-with-:~$ cd /challenge/files
hacker@globbing~matching-with-:/challenge/files$ /challenge/run file_[bash]
You got it! Here is your flag!
pwn.college{kUh2nnpwh9Bhnowmwut6sgGeynB.QXzID00wyN5QTMxIzW}
hacker@globbing~matching-with-:/challenge/files$
```

🎉 You've already solved Matching with []! 🎉

>_ Terminal

</> Code

Desktop

SSH

ynB.QXzID00wyN5QTMxIzW}

↺

□ sudo

4. Matching paths with []:

```
hacker@globbing-matching-paths-with-:~$ /challenge/run /challenge/files/file_[ba  
sh]  
You got it! Here is your flag!  
pwn.college{YeeejQUdntAnkabzg5BiWghMCSU.QX0ID00wyN5QTMxIzW}  
hacker@globbing-matching-paths-with-:~$
```

🎉 You've already solved Matching paths with []! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 ICSU.QX0ID00wyN5QTMxIzW}



📄 sudo

5. Multiple globs:

```
hacker@globbing-multiple-globs:~$ cd /challenge/files
hacker@globbing-multiple-globs:/challenge/files$ /challenge/run *p*
You got it! Here is your flag!
pwn.college{MLUhrblWpNfJH-RIin9NXi0HYgL.0IM3kjNxwyN5QTMxIzW}
hacker@globbing-multiple-globs:/challenge/files$
```

🎉 You've already solved Multiple globs! 🎉

>_ Terminal

</> Code

Desktop

SSH



YgL.OIM3kjNxwyN5QTMxIzW}



sudo

6. Mixing globs:

```
hacker@globbing-mixing-globs:~$ cd /challenge/files
hacker@globbing-mixing-globs:/challenge/files$ /challenge/run [cep]
Your expansion did not expand to the requested files (challenging, educational,
pwning). Instead, it expanded to:
[cep]
hacker@globbing-mixing-globs:/challenge/files$ /challenge/run [cep]*
You got it! Here is your flag!
pwn.college{spRckwN82ApKk12bN-D8ifEAnqZ.QX1ID00wyN5QTMxIzW}
hacker@globbing-mixing-globs:/challenge/files$
```

🎉 You've already solved Mixing globs! 🎉

>_ Terminal

</> Code

Desktop

SSH

🚩 AnqZ.QX1ID00wyN5QTMxIzW}

↻

📄 sudo

7. Exclusionary globbing:

```
hacker@globbing-exclusionary-globbing:~$ cd /challenge/files
hacker@globbing-exclusionary-globbing:/challenge/files$ /challenge/run [!pwn]*
You got it! Here is your flag!
pwn.college{YXBRRbwYlNwZLHawyBamQvCQV8P.QX2ID00wyN5QTMxIzW}
hacker@globbing-exclusionary-globbing:/challenge/files$
```

🎉 You've already solved Exclusionary globbing! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 pwn.college{YXBRRbwYlNwZLHawyBamQvCQV8P.QX2ID00wyN5QTMxIzW}



📄 sudo

8. Tab completion:

```
hacker@globbing~tab-completion:~$ cat /challenge/pwncollege
pwn.college{c85VJkXsK7sdIei0tks20qnVwRj.OFN0EzNxwyN5QTMxIzW}
hacker@globbing~tab-completion:~$
```

🎉 You've already solved Tab completion! 🎉

>_ Terminal

</> Code

Desktop

SSH

vRj.OFN0EzNxwyN5QTMxIzW}



sudo

9. Multiple options for tab completion:

```
hacker@globbering-multiple-options-for-tab-completion:~$ cat /challenge/files/pwn
pwn                                pwncollege-family          pwncollege-flyswatter
pwn-college                       pwncollege-flag            pwncollege-hacking
pwn-the-planet                   pwncollege-flamingo
hacker@globbering-multiple-options-for-tab-completion:~$ cat /challenge/files/pwn
No flag in this file!
hacker@globbering-multiple-options-for-tab-completion:~$ cat /challenge/files/pwn-
college
No flag in this file!
hacker@globbering-multiple-options-for-tab-completion:~$ cat /challenge/files/pwn-
the-planet
No flag in this file!
hacker@globbering-multiple-options-for-tab-completion:~$ cat /challenge/files/pwnc
ollege-family
No flag in this file!
hacker@globbering-multiple-options-for-tab-completion:~$ cat /challenge/files/pwnc
ollege-flag
pwn.college{QCwgaV4z_0FbasniyIr7LMVM1GR.0IN0EzNxwyN5QTMxIzW}
hacker@globbering-multiple-options-for-tab-completion:~$
```



You've already solved Multiple options for tab completion!



> Terminal

</> Code

Desktop

SSH



1GR.0IN0EzNxwyN5QTMxIzW}



sudo

10. Tab completion on commands:

```
hacker@globbing~tab-completion-on-commands:~$ pwncollege-4537
Correct! Here is your flag:
pwn.college{8VLxSukyZ9ebjET30h0ZsAwrsvX.0VN0EzNxwyN5QTMxIzW}
hacker@globbing~tab-completion-on-commands:~$
```



You've already solved Tab completion on commands!



>_ Terminal

</> Code

Desktop

SSH



vX.0VN0EzNxwyN5QTMxIzW}



sudo

6. Practice Piping:

1. Redirecting output:

```
You have created the COLLEGE file and written the right value to it, but it  
doesn't look like you did it via output redirection.  
hacker@piping-redirecting-output:~$ echo PwN > COLLEGE  
Correct! You successfully redirected 'PwN' to the file 'COLLEGE'! Here is your  
flag:  
pwn.college{kZiZoGhWbwY-r60Fx4yvHU_owZ.QX0YTN0wyN5QTMxIzW}  
hacker@piping-redirecting-output:~$
```

🎉 You've already solved Redirecting output! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 >WZ.QX0YTN0wyN5QTMxIzW}

↻

📄 sudo

2. Redirecting more output:

```
hacker@piping-redirecting-more-output:~$ echo /challenge/run > myflag
hacker@piping-redirecting-more-output:~$ /challenge/run > myflag
[INFO] WELCOME! This challenge makes the following asks of you:
[INFO] - the challenge will check that output is redirected to a specific file path : myflag
[INFO] - the challenge will output a reward file if all the tests pass : /flag

[HYPE] ONWARDS TO GREATNESS!

[INFO] This challenge will perform a bunch of checks.
[INFO] If you pass these checks, you will receive the /flag file.

[TEST] You should have redirected my stdout to a file called myflag. Checking...

[PASS] The file at the other end of my stdout looks okay!
[PASS] Success! You have satisfied all execution requirements.
hacker@piping-redirecting-more-output:~$ cat myflag

[FLAG] Here is your flag:
[FLAG] pwn.college{YSGtCSSnLDntKcE4twQcIS0nbr.QX1YTN0wyN5QTMxIzW}

hacker@piping-redirecting-more-output:~$
```

 You've already solved Redirecting more output!



 Terminal

 Code

 Desktop

 SSH

 nbr.QX1YTN0wyN5QTMxIzW}



 sudo

3. Appending output:

```
hacker@piping-appending-output:~$ /challenge/run >> /home/hacker/the-flag
[INFO] WELCOME! This challenge makes the following asks of you:
[INFO] - the challenge will check that output is redirected to a specific file path : /home/hacker/the-flag

[HYPE] ONWARDS TO GREATNESS!

[INFO] This challenge will perform a bunch of checks.
[INFO] Good luck!

[TEST] You should have redirected my stdout to a file called /home/hacker/the-flag. Checking...

[HINT] File descriptors are inherited from the parent, unless the FD_CLOEXEC is set by the parent on the file descriptor.
[HINT] For security reasons, some programs, such as python, do this by default in certain cases. Be careful if you are
[HINT] creating and trying to pass in FDs in python.

[PASS] The file at the other end of my stdout looks okay!
[PASS] Success! You have satisfied all execution requirements.
I will write the flag in two parts to the file /home/hacker/the-flag! I'll do the first write directly to the file, and the second write, I'll do to stdout (if it's pointing at the file). If you redirect the output in append mode, the second write will append to (rather than overwrite) the first write, and you'll get the whole flag!
hacker@piping-appending-output:~$ cat /home/hacker/the-flag
|
\|/ This is the first half:
v
pwn.college{I8V6xs506LmhSBiW6qF8cQ1eKHq.QX3AT00wyN5QTMxIzW}
      ^
      that is the second half /|\
      |

If you only see the second half above, you redirected in *truncate* mode (>) rather than *append* mode (>>), and so the write of the second half to stdout overwrote the initial write of the first half directly to the file. Try append mode!
```

hacker@piping-appending-output:~\$

🎉 You've already solved Appending output! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 KHq.QX3AT00wyN5QTMxIzW}

↺

📄 sudo

4. Redirecting errors:

```
hacker@piping-redirecting-errors:~$ /challenge/run > myflag > instructions
[INFO] WELCOME! This challenge makes the following asks of you:
[INFO] - the challenge will check that output is redirected to a specific file path : myflag
[INFO] - the challenge will check that error output is redirected to a specific file path : instructions
[INFO] - the challenge will output a reward file if all the tests pass : /flag

[HYPE] ONWARDS TO GREATNESS!

[INFO] This challenge will perform a bunch of checks.
[INFO] If you pass these checks, you will receive the /flag file.

[TEST] You should have redirected my stdout to a file called myflag. Checking...

[FAIL] You did not satisfy all the execution requirements.
[FAIL] Specifically, you must fix the following issue:
[FAIL] You have redirected the wrong file for stdout (/home/hacker/instructions instead of myflag).
hacker@piping-redirecting-errors:~$ /challenge/run > myflag 2> instructions
hacker@piping-redirecting-errors:~$ cat /myflag
cat: /myflag: No such file or directory
hacker@piping-redirecting-errors:~$ cat myflag

[FLAG] Here is your flag:
[FLAG] pwn.college{oEGydQdlzdkZHanaAp17vpHl2dT.QX3YTN0wyN5QTMxIzW}

hacker@piping-redirecting-errors:~$
```



Successfully completed Redirecting errors!



> Terminal

</> Code

Desktop

SSH



2dT.QX3YTN0wyN5QTMxIzW}



sudo

5. Redirecting input:

```
hacker@piping-redirecting-input:~$ echo COLLEGE > PWN
hacker@piping-redirecting-input:~$ /challenge/run < PWN
Reading from standard input...
Correct! You have redirected the PWN file into my standard input, and I read
the value 'COLLEGE' out of it!
Here is your flag:
pwn.college{UjfoBm-In_FRQ1h-_sb29E0lf95.QXwcTN0wyN5QTMxIzW}
hacker@piping-redirecting-input:~$
```

🎉 Successfully completed Redirecting input! 🎉

>_ Terminal

</> Code

Desktop

SSH

f95.QXwcTN0wyN5QTMxIzW}



sudo

6. Grepping stored results:

```
hacker@piping-grepping-stored-results:~$ /challenge/run > /tmp/data.txt
[INFO] WELCOME! This challenge makes the following asks of you:
[INFO] - the challenge will check that output is redirected to a specific file path : /tmp/data.txt
[INFO] - the challenge will output a reward file if all the tests pass : /challenge/.data.txt

[HYPE] ONWARDS TO GREATNESS!

[INFO] This challenge will perform a bunch of checks.
[INFO] If you pass these checks, you will receive the /challenge/.data.txt file.

[TEST] You should have redirected my stdout to a file called /tmp/data.txt. Checking...

[HINT] File descriptors are inherited from the parent, unless the FD_CLOEXEC is set by the parent on the file descriptor.
[HINT] For security reasons, some programs, such as python, do this by default in certain cases. Be careful if you are
[HINT] creating and trying to pass in FDs in python.

[PASS] The file at the other end of my stdout looks okay!
[PASS] Success! You have satisfied all execution requirements.
hacker@piping-grepping-stored-results:~$ grep pwn.college /tmp/data.txt
pwn.college{UdhjF5THTLeBY2WtTrzRPh7060M.QX4ED00wyN5QTMxIzW}
hacker@piping-grepping-stored-results:~$
```

 Successfully completed Grepping stored results! 

>_ Terminal 

</> Code 

Desktop 

SSH 

 }OM.QX4ED00wyN5QTMxIzW}



 sudo

7. Grepping live output:

```
hacker@piping-grepping-live-output:~$ /challenge/run | grep pwn.college
[INFO] WELCOME! This challenge makes the following asks of you:
[INFO] - the challenge checks for a specific process at the other end of stdout
: grep
[INFO] - the challenge will output a reward file if all the tests pass : /challenge/.data.txt

[HYPE] ONWARDS TO GREATNESS!

[INFO] This challenge will perform a bunch of checks.
[INFO] If you pass these checks, you will receive the /challenge/.data.txt file.

[TEST] You should have redirected my stdout to another process. Checking...
[TEST] Performing checks on that process!

[INFO] The process' executable is /nix/store/sacz532zgiacvg7mva9v6gbfmyw427i3-gnugrep-3.12/bin/grep.
[INFO] This might be different than expected because of symbolic links (for example, from /usr/bin/python to /usr/bin/python3 to /usr/bin/python3.8).
[INFO] To pass the checks, the executable must be grep.

[PASS] You have passed the checks on the process on the other end of my stdout!
[PASS] Success! You have satisfied all execution requirements.
pwn.college{MxVLE7j92PrUcUfl2XPK6H6fpMt.QX5ED00wyN5QTMxIzW}
hacker@piping-grepping-live-output:~$
```

🎉 Successfully completed Grepping live output!



>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 5Mt.QX5ED00wyN5QTMxIzW}



📄 sudo

8. Grepping errors:

```
hacker@piping-grepping-errors:~$ /challenge/run 2>&1 | grep pwn.college
[INFO] WELCOME! This challenge makes the following asks of you:
[INFO] - the challenge checks for a specific process at the other end of stderr
: grep
[INFO] - the challenge will output a reward file if all the tests pass : /challenge/.data.txt

[HYPE] ONWARDS TO GREATNESS!

[INFO] This challenge will perform a bunch of checks.
[INFO] If you pass these checks, you will receive the /challenge/.data.txt file.

[TEST] You should have redirected my stderr to another process. Checking...
[TEST] Performing checks on that process!

[INFO] The process' executable is /nix/store/sacz532zgicvq7mva9v6gbfmyw427i3-gnugrep-3.12/bin/grep.
[INFO] This might be different than expected because of symbolic links (for example, from /usr/bin/python to /usr/bin/python3 to /usr/bin/python3.8).
[INFO] To pass the checks, the executable must be grep.

[PASS] You have passed the checks on the process on the other end of my stderr!
[PASS] Success! You have satisfied all execution requirements.
pwn.college{camTRUqBFIfHIYkKmz2pzsPANrZ.QX1AT00wyN5QTMxIzW}
hacker@piping-grepping-errors:~$
```

🎉 Successfully completed Grepping errors! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 ANrZ.QX1AT00wyN5QTMxIzW}



sudo

9. Filtering with grep -v:

```
hacker@piping-filtering-with-grep-v:~$ /challenge/run | grep -v DECOY
pwn.college{gHF3iC-zPLBCcw5UVYaKt6Bemmm.0FOxEzNxwyN5QTMxIzW}
hacker@piping-filtering-with-grep-v:~$
```

🎉 Successfully completed Filtering with grep -v!



>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 nm.0FOxEzNxwyN5QTMxIzW}



📄 sudo

10. Filtering with sed:

```
hacker@piping-filtering-with-sed:~$ /challenge/run | sed "s/FAKEFLAG//g"  
pwn.college{0Pfxcw0-DGNBM_Drf6m2s75JmHf.01NxQTMwyN5QTMxIzW}hacker@piping~filter  
ing-with-sed:~$
```

🎉 Successfully completed Filtering with sed! 🎉

>_ Terminal

</> Code

🖥️ Desktop

🔑 SSH

🚩 iHf.01NxQTMwyN5QTMxIzW}

↻

📄 sudo

11. Duplicating piped data with tee:

```
hacker@piping-duplicating-piped-data-with-tee:~$ /challenge/pwn | tee intercepted | /challenge/college
Processing...
WARNING: you are overwriting file intercepted with tee's output...
The input to 'college' does not contain the correct secret code! This code should be provided by the 'pwn' command. HINT: use 'tee' to save the first attempt to a file, inspect the saved output, and then retry the pipeline with what you learned.
hacker@piping-duplicating-piped-data-with-tee:~$ cat intercepted
Usage: /challenge/pwn --secret [SECRET_ARG]

SECRET_ARG should be "Q6wm5UiZ"
hacker@piping-duplicating-piped-data-with-tee:~$ /challenge/pwn --secret Q6wm5UiZ | /challenge/college
Processing...
Correct! Passing secret value to /challenge/college...
Great job! Here is your flag:
pwn.college{Q6wm5UiZA4YTAgy5-x5zUNbS_ug.QXxIT00wyN5QTMxIzW}
hacker@piping-duplicating-piped-data-with-tee:~$
```



Successfully completed Duplicating piped data with tee!



>_ Terminal

</> Code

Desktop

SSH



3_ug.QXxIT00wyN5QTMxIzW}



sudo

12. Process substitution for input:

```
hacker@piping~process-substitution-for-input:~$ diff <(/challenge/print_decoys)
<(/challenge/print_decoys_and_flag)
93a94
> pwn.college{0mxPg562vzZBamN4ktXF6XBaH_Q.0lNwMD0xwyN5QTMxIzW}
hacker@piping~process-substitution-for-input:~$
```

🎉 Successfully completed Process substitution
for input! 🎉

> Terminal

</> Code

Desktop

SSH

_Q.0lNwMD0xwyN5QTMxIzW}



sudo

13. Writing to multiple programs:

```
hacker@piping-writing-to-multiple-programs:~$ /challenge/hack | tee >(/challenge
/the) >(/challenge/planet)
This secret data must directly and simultaneously make it to /challenge/the and
/challenge/planet. Don't try to copy-paste it; it changes too fast.
18483290042590926302
Congratulations, you have duplicated data into the input of two programs! Here
is your flag:
pwn.college{Ee29XuJswQqxDTp63UiH52lUpZu.QXwgDN1wyN5QTMxIzW}
hacker@piping-writing-to-multiple-programs:~$
```



Successfully completed Writing to multiple
programs!



>_ Terminal

</> Code

Desktop

SSH



pZu.QXwgDN1wyN5QTMxIzW}



sudo

14. Split-piping stderr and stdout:

```
hacker@piping-split-piping-stderr-and-stdout:~$ /challenge/hack 2> >(/challenge/
the) | /challenge/planet
Congratulations, you have learned a redirection technique that even experts
struggle with! Here is your flag:
pwn.college{c9DL6SMP-G_Y02_3s51wmZel90N.QXxQDM2wyN5QTMxIzW}
hacker@piping-split-piping-stderr-and-stdout:~$
```

🎉 Successfully completed Split-piping stderr and stdout! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 }0N.QXxQDM2wyN5QTMxIzW}



📄 sudo

15. Named pipes:

```
hacker@piping-named-pipes:~$ mkfifo /tmp/flag_fifo
hacker@piping-named-pipes:~$ /challenge/run > /tmp/flag_fifo & cat /tmp/flag_fifo
0
You're successfully redirecting /challenge/run to a FIFO at /tmp/flag_fifo!
Bash will now try to open the FIFO for writing, to pass it as the stdout of
/challenge/run. Recall that operations on FIFOs will *block* until both the
read side and the write side is open, so /challenge/run will not actually be
launched until you start reading from the FIFO!
[1] 126
You've correctly redirected /challenge/run's stdout to a FIFO at
/tmp/flag_fifo! Here is your flag:
pwn.college{0d_KpLOJq0DUfpRuole11ip5UHv.01MzMD0xwyN5QTMxIzW}
[1]+  Done                  /challenge/run > /tmp/flag_fifo
hacker@piping-named-pipes:~$
```

🎉 Successfully completed Named pipes! 🎉

_ Terminal

</> Code

Desktop

SSH

🚩 -lv.01MzMD0xwyN5QTMxIzW}

↻

□ sudo

7. Shell Variables:

1. Printing Variables:

```
hacker@variables~printing-variables:~$ echo $FLAG  
pwn.college{gsPBhmT7ItHEbAX70yPr0nI3qzq.QX3UTN0wyN5QTMxIzW}  
hacker@variables~printing-variables:~$
```

🎉 Successfully completed Printing Variables! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 qzq.QX3UTN0wyN5QTMxIzW}



sudo

2. Setting Variables:

You've set the PWN variable, but it does not look like the value is correct.
Make sure that the value is COLLEGE!

```
hacker@variables~setting-variables:~$ PWN=COLLEGE
```

You've set the PWN variable properly! As promised, here is the flag:

```
pwn.college{s8TS1R6BmbCx0GMxHVpk6C0yufW.QX5UTN0wyN5QTMxIzW}
```

```
hacker@variables~setting-variables:~$
```

🎉 Successfully completed Setting Variables! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 ufW.QX5UTN0wyN5QTMxIzW}



sudo

3. Multi-word Variables:

```
hacker@variables~multi-word-variables:~$ PWN="COLLEGE YEAH"  
You've set the PwN variable properly! As promised, here is the flag:  
pwn.college{Q_KEZa7KFFgEK8SZn0TS49rycuN.QXwYTN0wyN5QTMxIzW}  
hacker@variables~multi-word-variables:~$
```

🎉 Successfully completed Multi-word Variables!



>_ Terminal

</> Code

Desktop

SSH

🚩 pwn.college{Q_KEZa7KFFgEK8SZn0TS49rycuN.QXwYTN0wyN5QTMxIzW}



sudo

4. Exploring Variables:

```
hacker@variables~exporting-variables:~$ export PWN=COLLEGE
You've set the PWN variable to the proper value!
hacker@variables~exporting-variables:~$ COLLEGE=PWN
You've set the PWN variable to the proper value!
You've set the COLLEGE variable to the proper value!
hacker@variables~exporting-variables:~$ /challenge/run
CORRECT!
You have exported PWN=COLLEGE and set, but not exported, COLLEGE=PWN. Great
job! Here is your flag:
pwn.college{EQMxNZILfVYKIrnnLCFkR918LmN.QXyYTN0wyN5QTMxIzW}
You've set the PWN variable to the proper value!
You've set the COLLEGE variable to the proper value!
hacker@variables~exporting-variables:~$
```

🎉 Successfully completed Exporting Variables! 🎉

>_ Terminal

</> Code

Desktop

SSH

_mN.QXyYTN0wyN5QTMxIzW}

↺

sudo

5. Printing Exported Variables:

```
hacker@variables-printing-exported-variables:~$ env | grep FLAG  
FLAG=pwn.college{Ielj37rTYwrV2arIPefACw7VnEf.QX4UTN0wyN5QTMxIzW}  
hacker@variables-printing-exported-variables:~$
```

🎉 Successfully completed Printing Exported Variables! 🎉

> Terminal

</> Code

Desktop

SSH

🚩 'nEf.QX4UTN0wyN5QTMxIzW}

↻

□ sudo

6. Storing Command Output:

```
hacker@variables~storing-command-output:~$ PWN=$(./challenge/run)
Congratulations! You have read the flag into the PwN variable. Now print it out
and submit it!
hacker@variables~storing-command-output:~$ echo $PWN
pwn.college{MqKo_NbdjP1yquw4_h-8qmKr22n.QX1cDN1wyN5QTMxIzW}
hacker@variables~storing-command-output:~$
```

🎉 Successfully completed Storing Command Output! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 r22n.QX1cDN1wyN5QTMxIzW}

↻

📄 sudo

7. Reading Input:

```
hacker@variables~reading-input:~$ read PWN
COLLEGE
You've set the PWN variable properly! As promised, here is the flag:
pwn.college{MLk5MKJJqta_GfAVhb3_TKMwSRY.QX4cTN0wyN5QTMxIzW}
hacker@variables~reading-input:~$
```

🎉 Successfully completed Reading Input! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 SRY.QX4cTN0wyN5QTMxIzW}

↻

📄 sudo

8. Reading Files:

```
hacker@variables~reading-files:~$ read PWN < /challenge/read_me
You've set the PWN variable properly! As promised, here is the flag:
pwn.college{MxHVDqNq0Trdbae6pwg8GK3hhxI.QXwID00wyN5QTMxIzW}
hacker@variables~reading-files:~$
```

🎉 Successfully completed Reading Files! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 hxl.QXwID00wyN5QTMxIzW}

↻

📄 sudo

8. Data Manipulation:

1. Translating characters:

```
hacker@data~translating-characters:~$ /challenge/run
Your case-swapped flag:
PwN.COLLEGE{mpY6z0IU_3vwa_4_hlwzipZtQ3o.01mXeZnXWYn5qtmXiZw}

hacker@data~translating-characters:~$ /challenge/run | tr 'A-Za-z' 'a-zA-Z'
yOUR CASE-SWAPPED FLAG:
pwn.college{MPy6Zoiu_3VwA_4_HLwZIPzTq30.01MxEzNxwyN5QTMxIzW}
```

🎉 You've already solved Translating characters! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 30.01MxEzNxwyN5QTMxIzW}

↻

📄 sudo

2. Deleting characters:

```
hacker@data~deleting-characters:~$ /challenge/run
Your character-stuffed flag:
p%w%n%.^%c^o%l^%l^%eg%e%[^40%h^7^%eu%1%8%j^7K^%4^%cA^S^j^%r^%u^%v^M^%k^%Z^%o^%xi^
7^I^.^%0^%F^%N^%x^%E^%z^%N^%x^%w^%y^%N^5%Q^%TM^%x^%I^%z^%W^%j^%
hacker@data~deleting-characters:~$ /challenge/run | tr -d '^%'
Your character-stuffed flag:
pwn.college{40h7eu18j7K4cASjruvMkZoxi7I.0FNxEzNxwyN5QTMxIzW}
hacker@data~deleting-characters:~$
```

🎉 Successfully completed Deleting characters! 🎉

> Terminal

</> Code

Desktop

SSH

❏ xI7I.0FNxEzNxwyN5QTMxIzW}

↺

❏ sudo

3. Deleting newlines:

```
hacker@data~deleting-newlines:~$ /challenge/run
Your line-split flag:
p
wn
.
co
ll
ege
{
gv-n
u
H
K
s
Q7
Gd
Y
Z
G
Ss
fE
nR0
nRKXq
.
0VN
xEzNxw
y
N5QTM
x
I
z
W
}

hacker@data~deleting-newlines:~$ /challenge/run | tr -d "\n"
Your line-split flag: pwn.college{gv-nuHksQ7GdYZGSsfEnR0nRKXq.0VNxEzNxwyN5QTMxIz
W}hacker@data~deleting-newlines:~$
```

🎉 Successfully completed Deleting newlines! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 .Xq.OVNxEzNxwyN5QTMxIzW}

↻

📄 sudo

4. Extracting the first lines with head:

```
hacker@data-extracting-the-first-lines-with-head:~$ /challenge/pwn | head -n 7|  
/challenge/college  
Congratulations, you piped the right codes!  
pwn.college{ARrEp4eonaV-iuRXro8vNS16-AC.01NxEzNxwyN5QTMxIzW}  
hacker@data-extracting-the-first-lines-with-head:~$
```

🎉 Successfully completed Extracting the first
lines with head! 🎉

>_ Terminal  </> Code  Desktop  SSH 

🚩 -AC.01NxEzNxwyN5QTMxIzW}



 sudo

5. Extracting specific sections of text:

```
hacker@data-extracting-specific-sections-of-text:~$ /challenge/run
14663 p
20424 w
9587 n
16054 .
22705 c
22276 o
3339 l
24436 l
21219 e
26647 g
29001 e
29932 {
15217 E
16683 l
12972 f
288 b
13588 0
31653 F
8249 D
22569 u
6176 H
4009 y
26371 D
8607 k
28075 R
23878 Q
6202 I
2224 n
8224 w
28150 h
29596 -
7852 7
4976 U
16504 9
27928 v
10908 8
24404 f
29051 l
11051 h
4082 .
29354 0
```

```
25611 1
23813 N
8451 x
25372 E
13801 z
6137 N
28142 x
387 w
26431 y
20999 N
21329 5
20900 Q
10256 T
5361 M
19918 x
17918 I
29386 z
8190 W
22423 }
hacker@data-extracting-specific-sections-of-text:~$ /challenge/run | cut -d " "
-f 2 | tr -d "\n"
pwn.college{Elfb0FDuHyDkRQInwh-7U9v8f1h.01NxExNxwyN5QTMxIzW}hacker@data-extracti
ng-specific-sections-of-text:~$
```

>_ Terminal

</> Code

Desktop

SSH

{flh.01NxExNxwyN5QTMxIzW}

↺

□ sudo

6. Sorting data:

```
hacker@data~sorting-data:~$ sort /challenge/flags.txt | tail
pwn.college{Mrb0Dt1bxo0JzxwY9joJ8wzglhy.0FM0MD0xwyN5QTMxIzW}
pwn.college{Mrb0Dt2bxn0JzxwZ9jnK8wzglhy.0EM0MD0xwyN5QTMxIzW}
pwn.college{Mrb0Dt2bxn0JzxwZ8joK8wzglhy.0FM0MD0xwyN5QTMxHzW}
pwn.college{Mrb0Dt2bxo0JzxwZ9joK7wzglhy.0FM0MD0xwxN5QTMxIzW}
pwn.college{Mrb0Dt2bxo0JzxwY9joJ8wzglhy.0FM0MD0xwyN4QTLxIzV}
pwn.college{Mrb0Dt2bxo0JzxwY9joK7wzgkxh.0EM0MD0xwyN5QTMxIzW}
pwn.college{Mrb0Dt2bxo0JzxwZ9joK8wzglhy.0FL0MD0xwyN5QTMxIzW}
pwn.college{Mrb0Dt2bxo0JzxwZ9joK8wzglhy.0FM0MC0xvyN5QTMxIzW}
pwn.college{Mrb0Dt2bxo0JzxwZ9joK8wzglhy.0FM0MD0xwyN5QTMxIyW}
pwn.college{Mrb0Dt2bxo0JzxwZ9joK8wzglhy.0FM0MD0xwyN5QTMxIzW}
hacker@data~sorting-data:~$
```

🎉 Successfully completed Sorting data! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 y.OFM0MD0xwyN5QTMxIzW}



📄 sudo

9. Processes and Jobs:

1. Listing Processes:

```
hacker@processes~listing-processes:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1         0  0  07:56 ?        00:00:00 /sbin/docker-init -- /nix/va
root           7         1  0  07:56 ?        00:00:00 /run/dojo/bin/sleep 6h
root          110         1  0  07:56 ?        00:00:00 /challenge/1028-run-32753
root          113        110  0  07:56 ?        00:00:00 sleep 6h
hacker         124         1  0  07:56 ?        00:00:00 /nix/store/nzhm2kbnlpp13kwwk
hacker         126        124  0  07:56 pts/0    00:00:00 /run/dojo/bin/bash --login
hacker         138        126  0  07:57 pts/0    00:00:00 ps -ef
hacker@processes~listing-processes:~$ /challenge/1028-run-32753
Yahaha, you found me! Here is your flag:
pwn.college{sHgSMMetx2xhBLwR8JG10xB0Uh5.QX4MD00wyN5QTMxIzW}
Now I will sleep for a while (so that you could find me with 'ps').
█
```

🎉 Successfully completed Listing Processes! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 Jh5.QX4MD00wyN5QTMxIzW}

↻

📄 sudo

2. Killing Processes:

```
hacker@processes~killing-processes:~$ ps -ef | grep dont_run
hacker      110      109  0 07:58 ?        00:00:00 /challenge/dont_run
hacker      133      124  0 08:00 pts/0    00:00:00 grep --color=auto dont_run
hacker@processes~killing-processes:~$ kill 133
bash: kill: (133) - No such process
hacker@processes~killing-processes:~$ kill 110
hacker@processes~killing-processes:~$ /challenge/run
Great job! Here is your payment:
pwn.college{wwoEMt83mK-Lbpb3chHAYEIe0UE.QXyQDO0wyN5QTMxIzW}
hacker@processes~killing-processes:~$
```

🎉 Successfully completed Killing Processes! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 JUE.QXyQDO0wyN5QTMxIzW}

↻

📄 sudo

3. Interrupting Processes:

```
hacker@processes~interrupting-processes:~$ /challenge/run
I could give you the flag... but I won't, until this process exits. Remember,
you can force me to exit with Ctrl-C. Try it now!
^C
Good job! You have used Ctrl-C to interrupt this process! Here is your flag:
pwn.college{kWaSVhSZsQbY7bZj0uvqzqt_NnC.QXzQD00wyN5QTMxIzW}
hacker@processes~interrupting-processes:~$
```

 Successfully completed Interrupting Processes!



 Terminal 

 Code 

 Desktop 

 SSH 

 NnC.QXzQD00wyN5QTMxIzW}



 sudo

4. Killing Misbehaving Processes:

```
hacker@processes-killing-misbehaving-processes:~$ ps -ef
UID          PID     PPID  C  STIME TTY          TIME CMD
root           1         0  0  10:56 ?           00:00:00 /sbin/docker-init -- /nix/var/nix/pro
files/dojo-workspace/bin/dojo-init /run/dojo
root           7         1  0  10:56 ?           00:00:00 /run/dojo/bin/sleep 6h
root          111         1  0  10:56 ?           00:00:00 sleep 6h
root          112         1  0  10:56 ?           00:00:00 sleep 6h
root          113         1  0  10:56 ?           00:00:00 su -c exec /challenge/decoy > /tmp/fl
ag_fifo hacker
hacker        114        113  0  10:56 ?           00:00:00 /usr/bin/python3 /challenge/decoy
hacker        125         1  0  10:56 ?           00:00:00 /nix/store/nzhm2kbnlpp13kwvk192825y8p
ykhiag-ttyd-1.7.7/bin/ttyd --port 7681 --int
hacker        127        125  0  10:56 pts/0       00:00:00 /run/dojo/bin/bash --login
hacker        132        127  0  10:56 pts/0       00:00:00 ps -ef
hacker@processes-killing-misbehaving-processes:~$ kill 114
hacker@processes-killing-misbehaving-processes:~$ cat /tmp/flag_fifo & /challenge/run
[1] 133
pwn.college{KDR52bX7GbGwBWRcoweTJP7DVkeXxlWCeT00HTciv4s.vP0}
pwn.college{Gg1a.JASer3Hhe12BRtuPCujNYIsm12hRrqlT8ZFeabtd1R}
pwn.college{S.CvPGHgVrT0K0WH1W4Gjp2EU7IUYPL5l4u-fzl.Lfmmusn}
pwn.college{7kiWb.bwXnSegZ2r-9QkT38D0slgw9l4u9LPgA0Y0P3QnKH}
pwn.college{8nh.yjqYqhfmyGud54CRQ-1a3QQqX6.cuQeXzyKAdNFa5DM}
pwn.college{5dyVPG-00Zhs0IYxTMMd2C6bIvgPrzEcLV15tR77G2unXuG}
pwn.college{oZx5-z4bL.7uttG6qPf60UJvxCuhTwCjVpo2x7JQ9LtyWrK}
pwn.college{nu1J0LDLBiPMqHHB3bZ1sgegowA0lMwN6x03IEg5jw7EvUM}
pwn.college{bLDzeh3FFpwYTiYuaZoXe2CvbF2em6K8E8do0bFXLCqZ5lU}
pwn.college{tMXJYm6eDVByHSHlkt.V0AC2NaiJwqA0mvYF70giJZZBwI}
pwn.college{H--R30gPGe5jw8YpLawv0URUX0f1bP9v7.T.Ki94tnZV0za}
pwn.college{K-Bk2IILKciACcLth-WaHtuCPMp2lWyYcIvMxG0j-L8oVu5}
pwn.college{hQm12Cw0wddayCX1P-zr-kwSryI1IMGPN3k49WxkVEz8YY5}
pwn.college{.9pGfeyFy3M06EyW1sJrUGN23M40A2dGHNlpi73RZyGt7EwU}
pwn.college{nZJt8rwK2XRcgQMH1tCweTjMrS9kKFpNUHLH7t4gxauc4Yc}
pwn.college{MErwBJejsigYm5jFIqWQfnPxxP3DnZd2mP4yiktzN63hyNj}
pwn.college{3ptkgV-qKhEcpXUzzLGJbpFuS.1Bblq0iDq3F.1JmKwXIFN}
pwn.college{9K0gbsfCkrYsu-jVL0XhaZv.JKwG.Fh3UrpU7oLSiWYxrvu}
pwn.college{r90B3KftADLTjfe6SvE4wgSoSU6ggHqERiUXpdLnPw.10T0}
pwn.college{zJd3unbcyUe37JxRyYY2Dh5HfRbF0bmmK4-rRTKuTkvmME-}
pwn.college{2J802-ct8kF5tSbw.e.lidUU6y.t5k7F1P190fjHqCXc1PL}
pwn.college{ZCWEY0N1mwF7gTArJuZgrxd9FSAUzjyD22dAuYvkHgvJvc3}
pwn.college{9otKJoFs9ANmjT7vYF0sai6G9v8JxChZ5Ka3qkoci.GjkF8}
pwn.college{6CqsKARYoE3FXJ72Yxh6-CT3oqTpbn0K0xS7tgjLYPfgDJd}
pwn.college{-uVwznzI2JM9NmxcIyflyGXxDnQx1.CDe2C1H5xwjudJc4I}
```

```
pwn.college{Aoi8b6iAvFAwaqWSPW5N5wqRjrqlceNt1.tZG4p-i.FgGH0}  
pwn.college{20gD93VBoTe89JLSATk1FJJk6.gptcyTFrHKwGk8xG0xDC8}  
pwn.college{pcTztajQZ6gybUtXu-pvnBmqV-48pNGtt4-CZ8GpbGetEY7}  
pwn.college{skvVSF068Ql0pqJizVbBlG8nKbBSfBRSqZqhB5CS0ngbwJf}  
pwn.college{kYqAz5fsoT0yr9QMdBCT7KpLbB2t8A8l79z45W-MUpVl26m}  
pwn.college{cSNSaJIRb7DX4ogJe-aH6Y.0kjnv4tSZZ4dZ0-6PhSAY7j6}  
pwn.college{wbiLL5Rw0Fwv.NxSg6v6puhtfhpHaGo9dcv8t9sg.XSYd5e}  
pwn.college{-b0.Gd3VB0o0Qm1av7271auko-nfMTrv3anLPPr38wh2K70}  
pwn.college{j0NePohjyLMzkoJco9TIwBN4ZNM0uD7mbQAo1rY02pD.SUr}  
pwn.college{EQZmmrUNZ.pNQg8tKXXBBY5AeZ2IrhaGnKVbck4NrWwY2F3}  
pwn.college{QeD6ku577rw5A415HU9QTHhMBdHLZMVoJlbIvbjjNUwCTqA}  
pwn.college{BRfCfiwDCZvbW2pyfp76yrgVEeReX7Cwtsa-0etVU-iM8-z}  
pwn.college{kAxfgqmNhdGEAwQNwB-6fnCtbTvHATg8RNgzTHVZrqc85a.}  
pwn.college{moQ4fAco9IXPV.KQx01lvn3sso00H8hm8GStlpbRK045aRv}  
pwn.college{ei90rE-joThXT853o4zvfWsxRcb6f-upXdFBbuUNYFyZ0oZ}  
pwn.college{E4xqawLXLJPJ0wF9qI3g0Vp-Zc-yDvN5MW0f--eIp8E-q6S}  
pwn.college{znNUcnPfmkR71Q9blmsd5oYrn.qvLEuZDVs-wmW9P6JOT08}  
pwn.college{L26yDBBxqwnhBL2rS7GVJ5PwWinTXQVmKbCiEDpf2W0.29B}  
pwn.college{vbQSyhZ59idbGCS7Z4YqeV0Yz27c9R23u8S1USTtuCvvc.6}  
pwn.college{-iFagaraHxTzHVxxHHw9VArhf66Q4.3SI2t0A0muREDjNgf}  
pwn.college{ge.UU9g6fPeHN3EPxv0QgGo9Gv51iB9sVnSeaXcS6p5UFG6}  
pwn.college{Kiu4ikgZEifs4TmcDyArU0AzRgvXdl-e.ovQe9e6LV2Zl6i}  
pwn.college{VDLuGfoueBCsFJuCjeup5TS00HPTobQ3d659TrmMiUNA8c1}  
pwn.college{6TiP97h2VJG-gEbXXii3eyVfo2FPm0eHwYuo2F0GtwARNHl}  
pwn.college{SpmFFpYymIghdt6ynpMT.kUAS4RGzioozsdb4YY62l1FyB}  
pwn.college{qVzGw84Z-NJu0bGk140CyuAiEDNTKITZU7la5S5pPWeDR0h}  
pwn.college{BjIqKwsV2tUGnlCA4dpLeaIn6t4AxvCazQpoLKQneKFl-jT}  
pwn.college{cUy.AuL6UC.3sMqvbww7EAVBX3MZvjGwMP3mZY7vwm0AZ9d}  
pwn.college{UoLZGAyo7lxeTkENHsEcn9JEEjK0ucAYKenZM0MmXf5ktp6}  
pwn.college{M1AeS4pd4J1NNMG.q7hHFvaPRwBvacawDeshn30p5I-A1V4}  
pwn.college{jT7UM8z3FhVjw9.paIVHBeLk1ULEMm3McbDBpQ0-m42Bl.y}  
pwn.college{FUqy0yoxnRRQmQ0p9IAk1mBDu7KKyZrTI4GR5QkwRmdw2dl}  
pwn.college{ZtbWpzHdL60bm0MZMwlnomlvuZ21s8nTwAuQwzWuYwRJo0e}  
pwn.college{Bbed0FdQlwwV3Wmayu3woh1lun1NHhsVzDXjm.nam8nPPvL}  
pwn.college{SdF91BNBRcr8H23sfuokRg6JgAIn1kW1XZKvslzPuxfCYfE}  
pwn.college{LugsynH2n1bppCRBCMT72CC7pQ-cE5HpCtGFGJjLF1XJlb4}  
pwn.college{QUJbs-LKi0VfK8GRXjnVvK8X7Apw0ifzPc1To8FPr5KkAjiT}  
pwn.college{ML4F.CSrPe4m-jeJ3FkS4pyC5ztcGPwr6sFewcRxt9MVUBM}  
pwn.college{DwL-RcTFQq1jxEenumPlmPJxOxiX.kMlSNsiwddgYc2sw7K}  
Sending the flag to /tmp/flag_fifo!  
pwn.college{IhIjdx0ImyR8kdwkaJmaKyK1djJ.0FNzMD0xwyN5QTMxIzW}
```

hacker@processes-killing-misbehaving-processes:~\$

_ Terminal

</> Code

Desktop

SSH

!jJ.0FNzMD0xwyN5QTMxIzW}

↺

□ sudo

5. Suspending Processes:

```
hacker@processes~suspending-processes:~$ /challenge/run
I'll only give you the flag if there's already another copy of me running in
this terminal... Let's check!

UID          PID     PPID  C  STIME TTY          TIME CMD
root         120      115   0  10:59 pts/0        00:00:00 /bin/bash -p /challenge/run
root         122      120   0  10:59 pts/0        00:00:00 ps -f

I don't see a second me!

To pass this level, you need to suspend me and launch me again! You can
background me with Ctrl-Z or, if you're not ready to do that for whatever
reason, just hit Enter and I'll exit!
^Z
[1]+  Stopped                  /challenge/run
hacker@processes~suspending-processes:~$ /challenge/run
I'll only give you the flag if there's already another copy of me running in
this terminal... Let's check!

UID          PID     PPID  C  STIME TTY          TIME CMD
root         120      115   0  10:59 pts/0        00:00:00 /bin/bash -p /challenge/run
root         127      115   0  10:59 pts/0        00:00:00 /bin/bash -p /challenge/run
root         129      127   0  10:59 pts/0        00:00:00 ps -f

Yay, I found another version of me! Here is the flag:
pwn.college{kGr0Kv3aWedMRjFzn0VzbrIDyxz.QX1QD00wyN5QTMxIzW}
hacker@processes~suspending-processes:~$
```



Successfully completed Suspending Processes!



_ Terminal

</> Code

Desktop

SSH

lyxz.QX1QD00wyN5QTMxIzW}



sudo

6. Resuming Processes:

```
hacker@processes~resuming-processes:~$ /challenge/run
Let's practice resuming processes! Suspend me with Ctrl-Z, then resume me with
the 'fg' command! Or just press Enter to quit me!
^Z
[1]+  Stopped                  /challenge/run
hacker@processes~resuming-processes:~$ fg
/challenge/run
I'm back! Here's your flag:
pwn.college{A47yqwYJUs6szJvUNQBaj7b4pt7.QX2QD00wyN5QTMxIzW}
Don't forget to press Enter to quit me!

```

🎉 Successfully completed Resuming Processes!



>_ Terminal

</> Code

Desktop

SSH

.pt7.QX2QD00wyN5QTMxIzW}



sudo

7. Backgrounding Processes:

```
hacker@processes~backgrounding-processes:~$ /challenge/run
I'll only give you the flag if there's already another copy of me running *and
not suspended* in this terminal... Let's check!

UID          PID  STAT  CMD
root          120  S+    /bin/bash -p /challenge/run
root          122  R+    ps -o user=UID,pid,stat,cmd

I don't see a second me!

To pass this level, you need to suspend me, resume the suspended process in the
background, and then launch a new version of me! You can background me with
Ctrl-Z (and resume me in the background with 'bg') or, if you're not ready to
do that for whatever reason, just hit Enter and I'll exit!
^Z
[1]+  Stopped                  /challenge/run
hacker@processes~backgrounding-processes:~$ bg
[1]+ /challenge/run &
hacker@processes~backgrounding-processes:~$

Yay, I'm now running the background! Because of that, this text will probably
overlap weirdly with the shell prompt. Don't panic; just hit Enter a few times
to scroll this text out.

hacker@processes~backgrounding-processes:~$ /challenge/run
I'll only give you the flag if there's already another copy of me running *and
not suspended* in this terminal... Let's check!

UID          PID  STAT  CMD
root          120  S     /bin/bash -p /challenge/run
root          130  S     sleep 6h
root          131  S+    /bin/bash -p /challenge/run
root          133  R+    ps -o user=UID,pid,stat,cmd

Yay, I found another version of me running in the background! Here is the flag:
pwn.college{8U2frpwBXCsm8t0F1BbKcBUwDQN.QX3QD00wyN5QTMxIzW}
hacker@processes~backgrounding-processes:~$
```

🎉 Successfully completed Backgrounding Processes! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 JQN.QX3QD00wyN5QTMxIzW}

↺

📄 sudo

8. Foregrounding Processes:

```
hacker@processes~foregrounding-processes:~$ /challenge/run
To pass this level, you need to suspend me, resume the suspended process in the
background, and *then* foreground it without re-suspending it! You can
background me with Ctrl-Z (and resume me in the background with 'bg') or, if
you're not ready to do that for whatever reason, just hit Enter and I'll exit!
^Z
[1]+  Stopped                  /challenge/run
hacker@processes~foregrounding-processes:~$ bg
[1]+ /challenge/run &
hacker@processes~foregrounding-processes:~$

Yay, I'm now running the background! Because of that, this text will probably
overlap weirdly with the shell prompt. Don't panic; just hit Enter a few times
to scroll this text out. After that, resume me into the foreground with 'fg';
I'll wait.

hacker@processes~foregrounding-processes:~$ fg
/challenge/run
YES! Great job! I'm now running in the foreground. Hit Enter for your flag!

pwn.college{w8xMA1owswWEDBTj_E3-nqXF5no.QX4QD00wyN5QTMxIzW}
hacker@processes~foregrounding-processes:~$
```

🎉 Successfully completed Foregrounding Processes! 🎉

> Terminal

</> Code

Desktop

SSH

5no.QX4QD00wyN5QTMxIzW}



sudo

9. Starting Backgrounded Processes:

```
hacker@processes~starting-backgrounded-processes:~$ /challenge/run
You've started me in the foreground! You must start me in the background (by
appending '&' to the command) to get the flag!
hacker@processes~starting-backgrounded-processes:~$ /challenge/run & sleep
[1] 123
sleep: missing operand
Try 'sleep --help' for more information.
hacker@processes~starting-backgrounded-processes:~$

Yay, you started me in the background! Because of that, this text will probably
overlap weirdly with the shell prompt, but you're used to that by now...

Anyways! Here is your flag!
pwn.college{4S0REENaFLYXD6uTng0DrfN5Ftx.QX5QD00wyN5QTMxIzW}
█
```

🎉 Successfully completed Starting Backgrounded Processes! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 Ftx.QX5QD00wyN5QTMxIzW}

↻

📄 sudo

10. Process Exit Codes:

```
hacker@processes~process-exit-codes:~$ /challenge/get-code
Exiting with an error code!
hacker@processes~process-exit-codes:~$ echo $?
164
hacker@processes~process-exit-codes:~$ /challenge/submit-code
You must run /challenge/submit-code with the exit code you retrieved from
/challenge/get-code as the first argument:

Usage: /challenge/submit-code [EXIT_CODE]
hacker@processes~process-exit-codes:~$ /challenge/submit-code 164
CORRECT! Here is your flag:
pwn.college{0RCwg_5Q0bp5whej8pLAIpFgorQ.QX5YD01wyN5QTMxIzW}
hacker@processes~process-exit-codes:~$
```

🎉 Successfully completed Process Exit Codes! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 jorQ.QX5YD01wyN5QTMxIzW}



📄 sudo

10. Untangling Users:

1. Becoming root with su:

```
hacker@users~becoming-root-with-su:~$ su
Password:
root@users~becoming-root-with-su:/home/hacker# ls
COLLEGE  a          instructions  my          myflag2      the-flag
Desktop  errors.log intercepted    myflag      not-the-flag
PWN      exec-suid  leap         myflag.2    output.log
root@users~becoming-root-with-su:/home/hacker# cat myflag

[FLAG] Here is your flag:
[FLAG] pwn.college{oEGydQdlzdkZHanaAp17vpHl2dT.QX3YTN0wyN5QTMxIzW}

root@users~becoming-root-with-su:/home/hacker# ls not-the-flag
not-the-flag
root@users~becoming-root-with-su:/home/hacker# cat not-the-flag
pwn.college{MsiwDL-8aCdGTesZe4VLCTmoyYG.QX1UDN1wyN5QTMxIzW}
root@users~becoming-root-with-su:/home/hacker#
```



Successfully completed Becoming root with su!



>_ Terminal

</> Code

Desktop

SSH

yxYG.QX1UDN1wyN5QTMxIzW}



sudo

2. Other users with su:

```
hacker@users~other-users-with-su:~$ su zardus
Password:
zardus@users~other-users-with-su:/home/hacker$ /challenge/run
Congratulations, you have become Zardus! Here is your flag:
pwn.college{UhiUnDNVb0wt2-j716hkRSx9Bq4.QX2UDN1wyN5QTMxIzW}
zardus@users~other-users-with-su:/home/hacker$
```

🎉 Successfully completed Other users with su! 🎉

_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 Bq4.QX2UDN1wyN5QTMxIzW}

↻

📄 sudo

3. Cracking passwords:

```
hacker@users~cracking-passwords:~$ john /challenge/shadow-leak
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Will run 40 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
aardvark          (zardus)
1g 0:00:00:13 100% 2/3 0.07639g/s 449.6p/s 449.6c/s 449.6C/s Johnson..buzz
Use the "--show" option to display all of the cracked passwords reliably
Session completed
hacker@users~cracking-passwords:~$ su zardus
Password:
zardus@users~cracking-passwords:/home/hacker$ /challenge/run
Congratulations, you have become Zardus! Here is your flag:
pwn.college{INqI43mb36n48DvMD-jY30RhhwM.QX3UDN1wyN5QTMxIzW}
zardus@users~cracking-passwords:/home/hacker$
```

🎉 Successfully completed Cracking passwords! 🎉

> Terminal

</> Code

Desktop

SSH

🚩 pwn.college{INqI43mb36n48DvMD-jY30RhhwM.QX3UDN1wyN5QTMxIzW}

↻

📄 sudo

4. Using sudo:

```
hacker@users~using-sudo:~$ ls
COLLEGE  a      instructions  my      myflag2      the-flag
Desktop  errors.log  intercepted  myflag  not-the-flag  '~'
PWN      exec-suid  leap        myflag.2  output.log

hacker@users~using-sudo:~$ cat not-the-flag
cat: not-the-flag: Permission denied
hacker@users~using-sudo:~$ sudo cat not-the-flag
pwn.college{EhNp_bnlAVrb8YvoqWK-tYYjzgg.QX4UDN1wyN5QTMxIzW}
hacker@users~using-sudo:~$
```

🎉 Successfully completed Using sudo! 🎉

>_ Terminal

</> Code

Desktop

SSH

🚩 |zgg.QX4UDN1wyN5QTMxIzW}



sudo

11. Perceiving Permissions:

1. Changing File Ownership:

```
hacker@permissions~changing-file-ownership:~$ chown hacker /flag
hacker@permissions~changing-file-ownership:~$ cat /flag
pwn.college{osh1grDyrgfbKIruoVivXIE9yQW.QXxEjN0wyN5QTMxIzW}
hacker@permissions~changing-file-ownership:~$
```



Successfully completed Changing File
Ownership!



>_ Terminal

</> Code

Desktop

SSH

lyQW.QXxEjN0wyN5QTMxIzW}



sudo

2. Group and Files:

```
hacker@permissions~groups-and-files:~$ chgrp hacker /flag
hacker@permissions~groups-and-files:~$ cat /flag
pwn.college{AB4jgcNU_0YqP-HKPltSvNJUhJ1.QXxcjM1wyN5QTMxIzW}
hacker@permissions~groups-and-files:~$
```

🎉 Successfully completed Groups and Files! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 UhJ1.QXxcjM1wyN5QTMxIzW}

↻

📄 sudo

3. Fun with Groups Names:

```
hacker@permissions~fun-with-groups-names:~$ id
uid=1000(hacker) gid=1000(grp31669) groups=1000(grp31669)
hacker@permissions~fun-with-groups-names:~$ chgrp grp31669 /flag
hacker@permissions~fun-with-groups-names:~$ cat /flag
pwn.college{QH0t5LX7egot4hlySwr3f8XbVSS.QXycjM1wyN5QTMxIzW}
hacker@permissions~fun-with-groups-names:~$
```

🎉 Successfully completed Fun With Groups Names! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 >VSS.QXycjM1wyN5QTMxIzW}

↻

📄 sudo

4. Changing Permissions:

```
hacker@permissions~changing-permissions:~$ chmod a+r /flag  
hacker@permissions~changing-permissions:~$ cat /flag  
pwn.college{0e1fozKL37Z8pCnXpwEoH0Tz7Ky.QXzcjM1wyN5QTMxIzW}  
hacker@permissions~changing-permissions:~$
```

 Successfully completed Changing Permissions!



>_ Terminal 

</> Code 

Desktop 

SSH 

 z7Ky.QXzcjM1wyN5QTMxIzW}



 sudo

5. Executable Files:

```
hacker@permissions~executable-files:~$ chmod a+x /challenge/run
hacker@permissions~executable-files:~$ /challenge/run
Successful execution! Here is your flag:
pwn.college{4ZzyjLE58cgMKHJoyv4PnA1li06.QXyEjN0wyN5QTMxIzW}
hacker@permissions~executable-files:~$
```

🎉 Successfully completed Executable Files! 🎉

> Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 1li06.QXyEjN0wyN5QTMxIzW}

↻

📄 sudo


```
hacker@permissions~permission-tweaking-practice:~$ chmod g+w /challenge/pwn
```

You set the correct permissions!

You've solved all 8 rounds! I have changed the ownership of the /flag file so that you can 'chmod' it. You won't be able to read it until you make it readable with chmod!

Current permissions of "/flag": -----

- the user doesn't have read permissions
- the user doesn't have write permissions
- the user doesn't have execute permissions
- the group doesn't have read permissions
- the group doesn't have write permissions
- the group doesn't have execute permissions
- the world doesn't have read permissions
- the world doesn't have write permissions
- the world doesn't have execute permissions

```
hacker@permissions~permission-tweaking-practice:~$ chmod u+r /flag
```

```
hacker@permissions~permission-tweaking-practice:~$ cat /flag
```

```
pwn.college{k8Xr1i4YDOLb0trKaXYAHAG8G1b.QXwEjN0wyN5QTMxIzW}
```

```
hacker@permissions~permission-tweaking-practice:~$
```

>_ Terminal

</> Code

Desktop

SSH

3G1b.QXwEjN0wyN5QTMxIzW}

↺

□ sudo

7. Permissions Setting Practice:

```
hacker@permissions-permissions-setting-practice:~$ chmod u=rw,g=r,o=rwx /challenge/pwn
You set the correct permissions!
Round 2 of 8!
```

```
Current permissions of "/challenge/pwn": rw-r--rwx
* the user does have read permissions
* the user does have write permissions
- the user doesn't have execute permissions
* the group does have read permissions
- the group doesn't have write permissions
- the group doesn't have execute permissions
* the world does have read permissions
* the world does have write permissions
* the world does have execute permissions
```

```
Needed permissions of "/challenge/pwn": -wx-wxrw
- the user doesn't have read permissions
* the user does have write permissions
* the user does have execute permissions
- the group doesn't have read permissions
* the group does have write permissions
* the group does have execute permissions
* the world does have read permissions
* the world does have write permissions
* the world does have execute permissions
```

```
hacker@permissions-permissions-setting-practice:~$ chmod u=wx,g=wx,o=rwx /challenge/pwn
You set the correct permissions!
Round 3 of 8!
```

```
Current permissions of "/challenge/pwn": -wx-wxrw
- the user doesn't have read permissions
* the user does have write permissions
* the user does have execute permissions
- the group doesn't have read permissions
* the group does have write permissions
* the group does have execute permissions
* the world does have read permissions
* the world does have write permissions
* the world does have execute permissions
```

```
hacker@permissions-permissions-setting-practice:~$ chmod u=x,g=rw,o=rwx /challenge/pwn
You set the correct permissions!
Round 4 of 8!
```

```
Current permissions of "/challenge/pwn": --xrw-rwx
- the user doesn't have read permissions
- the user doesn't have write permissions
* the user does have execute permissions
* the group does have read permissions
* the group does have write permissions
- the group doesn't have execute permissions
* the world does have read permissions
* the world does have write permissions
* the world does have execute permissions
```

```
Needed permissions of "/challenge/pwn": rw-rwx--x
* the user does have read permissions
* the user does have write permissions
- the user doesn't have execute permissions
* the group does have read permissions
* the group does have write permissions
* the group does have execute permissions
- the world doesn't have read permissions
- the world doesn't have write permissions
* the world does have execute permissions
```

```
hacker@permissions-permissions-setting-practice:~$ chmod u=rw,g=rwx,o=x /challenge/pwn
You set the correct permissions!
Round 5 of 8!
```

```
Current permissions of "/challenge/pwn": rw-rwx--x
* the user does have read permissions
* the user does have write permissions
- the user doesn't have execute permissions
* the group does have read permissions
* the group does have write permissions
* the group does have execute permissions
- the world doesn't have read permissions
- the world doesn't have write permissions
* the world does have execute permissions
```

```
You set the correct permissions!
Round 6 of 8!
```

```
Current permissions of "/challenge/pwn": r--r---w-
* the user does have read permissions
- the user doesn't have write permissions
- the user doesn't have execute permissions
* the group does have read permissions
- the group doesn't have write permissions
- the group doesn't have execute permissions
- the world doesn't have read permissions
* the world does have write permissions
- the world doesn't have execute permissions
```

```
Needed permissions of "/challenge/pwn": r-xrw-wx
* the user does have read permissions
- the user doesn't have write permissions
* the user does have execute permissions
* the group does have read permissions
* the group does have write permissions
- the group doesn't have execute permissions
- the world doesn't have read permissions
* the world does have write permissions
* the world does have execute permissions
```

```
hacker@permissions-permissions-setting-practice:~$ chmod u=r,x,g=rw,o=wx /challenge/pwn
You set the correct permissions!
Round 7 of 8!
```

```
Current permissions of "/challenge/pwn": r-xrw-wx
* the user does have read permissions
- the user doesn't have write permissions
* the user does have execute permissions
* the group does have read permissions
* the group does have write permissions
- the group doesn't have execute permissions
- the world doesn't have read permissions
* the world does have write permissions
* the world does have execute permissions
```

```
Needed permissions of "/challenge/pwn": --x---wx
- the user doesn't have read permissions
```

```
pwn
You set the correct permissions!
Round 8 of 8!
```

```
Current permissions of "/challenge/pwn": --x---wx
- the user doesn't have read permissions
- the user doesn't have write permissions
* the user does have execute permissions
- the group doesn't have read permissions
- the group doesn't have write permissions
- the group doesn't have execute permissions
- the world doesn't have read permissions
* the world does have write permissions
* the world does have execute permissions
```

```
Needed permissions of "/challenge/pwn": rw-----w-
* the user does have read permissions
* the user does have write permissions
- the user doesn't have execute permissions
- the group doesn't have read permissions
- the group doesn't have write permissions
- the group doesn't have execute permissions
- the world doesn't have read permissions
* the world does have write permissions
- the world doesn't have execute permissions
```

```
hacker@permissions-permissions-setting-practice:~$ chmod u=rw,g=-w /challenge/pwn
You set the correct permissions!
You've solved all 8 rounds! I have changed the ownership of the /flag file so that you can 'chmod' it. You won't be able to read it until you make it readable with chmod!
```

```
Current permissions of "/flag": -----
- the user doesn't have read permissions
- the user doesn't have write permissions
- the user doesn't have execute permissions
- the group doesn't have read permissions
- the group doesn't have write permissions
- the group doesn't have execute permissions
- the world doesn't have read permissions
- the world doesn't have write permissions
- the world doesn't have execute permissions
```

```
hacker@permissions~permissions-setting-practice:~$ chmod u=r /flag
hacker@permissions~permissions-setting-practice:~$ cat /flag
pwn.college{wFfT_siq5j60Lh16LumZaw22SyH.QXzET00wyN5QTMxIzW}
hacker@permissions~permissions-setting-practice:~$
```

[Terminal](#) [Code](#) [Desktop](#) [SSH](#)

[SyH.QXzET00wyN5QTMxIzW}](#) [sudo](#)

8. The SUID Bit:

```
hacker@permissions~the-suid-bit:~$ chmod u+s /challenge/getroot
hacker@permissions~the-suid-bit:~$ /challenge/getroot
SUCCESS! You have set the suid bit on this program, and it is running as root!
Here is your shell...
root@permissions~the-suid-bit:/home/hacker# cat /flag
pwn.college{UgYPHnPjTw14z50pkw5dtcABUtD.QXzEjN0wyN5QTMxIzW}
root@permissions~the-suid-bit:/home/hacker#
```

🎉 Successfully completed The SUID Bit! 🎉

[Terminal](#) [Code](#) [Desktop](#) [SSH](#)

[BUtD.QXzEjN0wyN5QTMxIzW}](#) [sudo](#)

12. Chaining Commands:

1. Chaining with Semicolons:

```
hacker@chaining~chaining-with-semicolons:~$ /challenge/pwn; /challenge/college
Yes! You chained /challenge/pwn and /challenge/college! Here is your flag:
pwn.college{40wXr3k5l57G2sZ1wkos_I2beSd.QX1UD00wyN5QTMxIzW}
hacker@chaining~chaining-with-semicolons:~$
```

🎉 Successfully completed Chaining with Semicolons! 🎉

> Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 eSd.QX1UD00wyN5QTMxIzW}

↻

📄 sudo

2. Building on Success:

```
hacker@chaining-building-on-success:~$ /challenge/first-success && /challenge/se  
cond  
Nice chaining! Flag: pwn.college{I4vf0WapkZJuIz1lRqlNzB4tvQp.0lM0MD0xwyN5QTMxIzW  
}  
hacker@chaining-building-on-success:~$
```

 Successfully completed Building on Success!



>_ Terminal 

</> Code 

Desktop 

SSH 

 }p.OlM0MD0xwyN5QTMxIzW}



 sudo

3. Handling Failure:

```
hacker@chaining-handling-failure:~$ /challenge/first-failure || /challenge/second
Nice chaining! Flag: pwn.college{YDqIqm919Bv-30IbqhlRybS79fD.01MOMDOxwyN5QTMxIzW}
hacker@chaining-handling-failure:~$
```

🎉 Successfully completed Handling Failure! 🎉

> Terminal

</> Code

Desktop

SSH

🚩 pwn.college{YDqIqm919Bv-30IbqhlRybS79fD.01MOMDOxwyN5QTMxIzW}



sudo

4. Your First Shell Script:

```
hacker@chaining-your-first-shell-script:~$ echo "/challenge/pwn">x.sh
hacker@chaining-your-first-shell-script:~$ echo "/challenge/college">>x.sh
hacker@chaining-your-first-shell-script:~$ bash x.sh
Great job, you've written your first shell script! Here is the flag:
pwn.college{sB4yYvITpkPZN6CYXUiAMCUMoLO.QXxcD00wyN5QTMxIzW}
hacker@chaining-your-first-shell-script:~$
```

 Successfully completed Your First Shell Script!


 Terminal 

 Code 

 Desktop 

 SSH 

 oLO.QXxcD00wyN5QTMxIzW}



 sudo

5. Redirecting Script Output:

```
hacker@chaining-redirecting-script-output:~$ /challenge/pwn
5dbbf06e67b514531b0c0b6ef2842159 -
hacker@chaining-redirecting-script-output:~$ /challenge/college
f944c933fb11e88fff6dae9058179866 -
hacker@chaining-redirecting-script-output:~$ bash x.sh | /challenge/solve
Correct! Here is your flag:
pwn.college{Yzzwp4M7cuiK85EPNj_Tw1mdvpm.QX4ET00wyN5QTMxIzW}
hacker@chaining-redirecting-script-output:~$
```



Successfully completed Redirecting Script
Output!



>_ Terminal

</> Code

Desktop

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wpm.QX4ET00wyN5QTMxIzW}



sudo

6. Executable Shell Scripts:

```
hacker@chaining-executable-shell-scripts:~$ echo "/challenge/solve">solve.sh
hacker@chaining-executable-shell-scripts:~$ chmod +x solve.sh
hacker@chaining-executable-shell-scripts:~$ ./solve.sh
Congratulations on your shell script execution! Your flag:
pwn.college{M9v3acn3Xwi23ACkvo-_aTuxI3X.QX0cjM1wyN5QTMxIzW}
hacker@chaining-executable-shell-scripts:~$
```



Successfully completed Executable Shell
Scripts! 🎉

>_ Terminal

</> Code

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xl3X.QX0cjM1wyN5QTMxIzW}



sudo

7. Understanding Shebangs:

```
hacker@chaining-understanding-shebangs:~$ nano /home/hacker/solve.sh
hacker@chaining-understanding-shebangs:~$ chmod +x /home/hacker/solve.sh
hacker@chaining-understanding-shebangs:~$ /challenge/run
Testing your script...
Perfect! Your flag:
Flag: pwn.college{wVNvzircbMeWl5-1b89Z2RdNH5t.0V0zMD0xwyN5QTMxIzW}

hacker@chaining-understanding-shebangs:~$
```

```
GNU nano 9.2 /home/hacker/solve.sh
#!/bin/bash
echo "hack the planet"
```

Successfully completed Understanding
Shebangs! 🎉

Help
Exit

Write Out
Read File

Where Is
Replace

Read 2 lines
Cut
Paste

Execute
Justify

Location
Go To Line

Terminal Code Desktop SSH

5t.0V0zMD0xwyN5QTMxIzW}

sudo

8. Scripting with Arguments:

```
hacker@chaining-scripting-with-arguments:~$ nano /home/hacker/solve.sh
hacker@chaining-scripting-with-arguments:~$ chmod a+x /home/hacker/solve.sh
hacker@chaining-scripting-with-arguments:~$ /home/hacker/solve.sh apple banana
banana apple
hacker@chaining-scripting-with-arguments:~$ /challenge/run
Correct! Your script properly reversed the arguments.
Here's your flag:
pwn.college{UT6SvLN0VYy2wGXhJMEhrLztsI8.0VNzMD0xwyN5QTMxIzW}
hacker@chaining-scripting-with-arguments:~$
```

🎉 Successfully completed Scripting with Arguments! 🎉

> _ Terminal </> Code 🖥 Desktop 🔑 SSH

🚩 I8.0VNzMD0xwyN5QTMxIzW}

```
GNU nano 9.2 /home/hacker/solve.sh
#!/bin/bash
echo "$2 $1"
```

⌘ Read 2 lines

⌘ Help	⌘ Write Out	⌘ Where Is	⌘ Cut	⌘ Execute	⌘ Location
⌘ Exit	⌘ Read File	⌘ Replace	⌘ Paste	⌘ Justify	⌘ Go To Line

9. Scripting with Conditionals:

```
hacker@chaining-scripting-with-conditionals:~$ nano /home/hacker/solve.sh
hacker@chaining-scripting-with-conditionals:~$ chmod a+x /home/hacker/solve.sh
hacker@chaining-scripting-with-conditionals:~$ /challenge/run
Correct! Your script properly handles all the conditions.
Here's your flag:
pwn.college{gJFvEI7FKtQtPo_zQJuVIfJi3PA.0lNzMD0xwyN5QTMxIzW}
hacker@chaining-scripting-with-conditionals:~$
```

🎉 Successfully completed Scripting with Conditionals! 🎉

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PA.0lNzMD0xwyN5QTMxIzW}

```
GNU nano 9.2 /home/hacker/solve.sh
#!/bin/bash
if [ "$1" == "pwn" ]
then
echo "college"
fi

[ Read 5 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

10. Scripting with Default Cases:

```
hacker@chaining-scripting-with-default-cases:~$ nano /home/hacker/solve.sh
hacker@chaining-scripting-with-default-cases:~$ chmod a+x /home/hacker/solve.sh
hacker@chaining-scripting-with-default-cases:~$ /challenge/run
Correct! Your script properly handles the if/else conditions.
Here's your flag:
pwn.college{Uw97N3it6FsJpxI3Cb8g3HUyKVl.01NzMDOxwyN5QTMxIzW}
hacker@chaining-scripting-with-default-cases:~$
```

🎉 Successfully completed Scripting with Default Cases! 🎉

>_ Terminal </> Code 🖥️ Desktop 🔑 SSH

🚩 {VI.01NzMDOxwyN5QTMxIzW}

```
GNU nano 9.2 /home/hacker/solve.sh
#!/bin/bash
if [ "$1" == "pwn" ]
then
echo "college"
else
echo "nope"
fi
```

[Read 7 lines]

Help	Write Out	Where Is	Cut	Execute	Location
Exit	Read File	Replace	Paste	Justify	Go To Line

11. Scripting with Multiple Conditions:

```
hacker@chaining-scripting-with-multiple-conditions:~$ nano /home/hacker/solve.sh
hacker@chaining-scripting-with-multiple-conditions:~$ chmod a+x /home/hacker/solve.sh
hacker@chaining-scripting-with-multiple-conditions:~$ /challenge/run
Correct! Your script properly handles all the conditions with elif.
Here's your flag:
pwn.college{w3TFkH3rNsFoI5in_84DQ5TCAeM.0F0zMD0xwyN5QTMxIzW}
hacker@chaining-scripting-with-multiple-conditions:~$
```

GNU nano 9.2 /home/hacker/solve.sh

```
#!/bin/bash
if [ "$1" == "hack" ]
then
echo "the planet"
elif [ "$1" == "pwn" ]
then
echo "college"
elif [ "$1" == "learn" ]
then
echo "linux"
else
echo "unknown"
fi
```

🎉 Successfully completed Scripting with Multiple Conditions! 🎉

➤ Terminal Code Desktop SSH

📁 :M.OF0zMD0xwyN5QTMxIzW}



[Read 13 lines]
^G Help ^G Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^N Replace ^U Paste ^J Justify ^_ Go To Line

12. Reading Shell Scripts:

```
hacker@chaining-reading-shell-scripts:~$ cat /challenge/run
#!/usr/bin/exec-suid -- /bin/bash -p

read GUESS
if [ "$GUESS" == "hack the PLANET" ]
then
    echo "CORRECT! Your flag:"
    cat /flag
else
    echo "Read the /challenge/run file to figure out the correct password!"
fi
hacker@chaining-reading-shell-scripts:~$ echo "hack the PLANET" | /challenge/run
CORRECT! Your flag:
pwn.college{MlmbzR00lEgXl6LkyPwCs6oIyoK.0lMwgD0xwyN5QTMxIzW}
hacker@chaining-reading-shell-scripts:~$
```



Successfully completed Reading Shell Scripts!



_ Terminal

</> Code

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SSH



oK.OlMwgD0xwyN5QTMxIzW}



sudo

13. Terminal Multiplexing:

1. Launching Screen:

```
GNU Screen version 4.09.01 (GNU) 20-Aug-23

Copyright (c) 2018-2023 Alexander Naumov, Amadeusz Slawinski
Copyright (c) 2015-2017 Juergen Weigert, Alexander Naumov, Amadeusz Slawinski
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Copyright (c) 2008-2009 Juergen Weigert, Michael Schroeder, Micah Cowan,
Sadrul Habib Chowdhury
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FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

You should have received a copy of the GNU General Public License along with
this program (see the file COPYING); if not, see
https://www.gnu.org/licenses/, or contact Free Software Foundation, Inc., 51
Franklin Street, Fifth Floor, Boston, MA 02111-1301 USA.

Send bugreports, fixes, enhancements, t-shirts, money, beer & pizza to
screen-devel@gnu.org

Capabilities:
+copy +remote-detach +power-detach +multi-attach +multi-user +font +color-256
+utf8 +rxvt +builtin-telnet

[Press Space or Return to end.]
```

```
Congratulations! You're inside a screen session!
Here's your flag:
pwn.college{QFbEqbQcwUTPjt2uljr8X9iTePV.0VN4ID0xwyN5QTMxIzW}
hacker@terminal-multiplexing-launching-screen:~$
```

Terminal

Code

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SSH

3PV.0VN4ID0xwyN5QTMxIzW}

Terminal

Code

Desktop

SSH

3PV.0VN4ID0xwyN5QTMxIzW}

sudo

 You've already solved Launching Screen! 

2. Detaching and Attaching:

```
hacker@terminal-multiplexing-detaching-and-attaching:~$ screen
[detached from 156.pts-0.terminal-multiplexing-detaching-and-attaching]
hacker@terminal-multiplexing-detaching-and-attaching:~$ /challenge/run
Found detached screen session: 156.pts-0.terminal-multiplexing-detaching-and-attaching
Sending flag to your screen session...
```

Flag sent! Now reattach to your screen session with:

```
screen -r
```

You'll find the flag waiting for you there!

```
hacker@terminal-multiplexing-detaching-and-attaching:~$ screen -r
[screen is terminating]
hacker@terminal-multiplexing-detaching-and-attaching:~$
```

🎉 Successfully completed Detaching and Attaching! 🎉

>_ Terminal [🔗](#) </> Code [🔗](#) 🖥️ Desktop [🔗](#) 🔑 SSH [🔗](#)

🚩 xDn.OIN4ID0xwyN5QTMxIzW}

```
hacker@terminal-multiplexing-detaching-and-attaching:~$ echo Yes! Flag is: pwn.college{YSjXnUYe3MyQRoJHx__DAo3ZxDn.01N4ID0xwyN5QTMxIzW}
Yes! Flag is: pwn.college{YSjXnUYe3MyQRoJHx__DAo3ZxDn.01N4ID0xwyN5QTMxIzW}
hacker@terminal-multiplexing-detaching-and-attaching:~$
```

🎉 Successfully completed Detaching and Attaching! 🎉

>_ Terminal [🔗](#) </> Code [🔗](#) 🖥️ Desktop [🔗](#) 🔑 SSH [🔗](#)

🚩 xDn.OIN4ID0xwyN5QTMxIzW}

🔄 sudo

3. Finding Sessions:

```
hacker@terminal-multiplexing-finding-sessions:~$ screen -ls
```

There are screens on:

123.session_5ca01f97933f52d0	(09/28/26 11:23:25)	(Detached)
129.session_aa4c238ab3fc148e	(09/28/26 11:23:25)	(Detached)
126.session_30ee179c8e872e97	(09/28/26 11:23:25)	(Detached)

3 Sockets in /run/screen/S-hacker.

```
hacker@terminal-multiplexing-finding-sessions:~$ screen -r 123
```

[detached from 123.session_5ca01f97933f52d0]

```
hacker@terminal-multiplexing-finding-sessions:~$
```



Successfully completed Finding Sessions!

```
hacker@terminal-multiplexing-finding-sessions:~$ echo 'Congratulations! You found the right session!'
```

Congratulations! You found the right session!

```
hacker@terminal-multiplexing-finding-sessions:~$ echo pwn.college{8b0dmHnSS6Ug9U_pXuHc024gBQz.01N4ID0xwyN5QTMxIzW}
```

```
pwn.college{8b0dmHnSS6Ug9U_pXuHc024gBQz.01N4ID0xwyN5QTMxIzW}
```

```
hacker@terminal-multiplexing-finding-sessions:~$
```



Successfully completed Finding Sessions! 🎉

>_ Terminal

</> Code

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SSH



BQz.01N4ID0xwyN5QTMxIzW}



sudo

4. Switching Windows:

```
cat <<MSG
Welcome to the window switching challenge!
You are currently in window 1.
The flag is hidden in window 0.
Use Ctrl-A 0 to switch to window 0!
MSG
hacker@terminal-multiplexing-switching-windows:~$ cat <<MSG
> Welcome to the window switching challenge!
> You are currently in window 1.
> The flag is hidden in window 0.
> Use Ctrl-A 0 to switch to window 0!
> MSG
Welcome to the window switching challenge!
You are currently in window 1.
The flag is hidden in window 0.
Use Ctrl-A 0 to switch to window 0!
hacker@terminal-multiplexing-switching-windows:~$
```

```
hacker@terminal-multiplexing-switching-windows:~$ cat <<MSG
> Excellent work! You found window 0!
> Here is your flag: pwn.college{A1jQ7pb4W-I3KvvKuQJgAJ4s82y.0F04ID0xwyN5QTMxIzW}
}
> MSG
Excellent work! You found window 0!
Here is your flag: pwn.college{A1jQ7pb4W-I3KvvKuQJgAJ4s82y.0F04ID0xwyN5QTMxIzW}
hacker@terminal-multiplexing-switching-windows:~$
```

🎉 Successfully completed Switching Windows! 🎉

> Terminal

</> Code

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SSH

32y.0F04ID0xwyN5QTMxIzW}

↻

sudo

5. Detaching and Attaching (tmux):

```
hacker@terminal-multiplexing-detaching-and-attaching-tmux:~$ tmux
[detached (from session 0)]
hacker@terminal-multiplexing-detaching-and-attaching-tmux:~$ /challenge/run
Found detached tmux session: 0
Sending flag to your tmux session...

Flag sent! Now reattach to your tmux session with:
tmux attach

You'll find the flag waiting for you there!
hacker@terminal-multiplexing-detaching-and-attaching-tmux:~$ tmux attach
[detached (from session 0)]
hacker@terminal-multiplexing-detaching-and-attaching-tmux:~$ █
```

🎉 Successfully completed Detaching and Attaching (tmux)! 🎉

```
hacker@terminal-multiplexing-detaching-and-attaching-tmux:~$ echo Congratulations, here
is your flag: pwn.college{c621Togn9FIzvnongpUGhiSN_vz.0V04ID0xwyN5QTMxIzW}
Congratulations, here is your flag: pwn.college{c621Togn9FIzvnongpUGhiSN_vz.0V04ID0xwyN5Q
TMxIzW}
hacker@terminal-multiplexing-detaching-and-attaching-tmux:~$ █
```

🎉 Successfully completed Detaching and Attaching (tmux)! 🎉

[0] 0:bas

Sep-26

>_ Terminal

</> Code

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_vz.0V04ID0xwyN5QTMxIzW}

↻

sudo

6. Switching Windows (tmux):

```
hacker@terminal-multiplexing-switching-windows-tmux:~$ cat <<MSG
> Excellent work! You found window 0!
> Here is your flag: pwn.college{gWyjE0rwDjAPmIADa6KAsXby1UE.0FM5ID0xwyN5QTMxIzW}
}
> MSG
Excellent work! You found window 0!
Here is your flag: pwn.college{gWyjE0rwDjAPmIADa6KAsXby1UE.0FM5ID0xwyN5QTMxIzW}
hacker@terminal-multiplexing-switching-windows-tmux:~$
```



Successfully completed Switching Windows
(tmux)!

>_ Terminal

</> Code

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IUE.0FM5ID0xwyN5QTMxIzW}



sudo

14. Pondering PATH:

1. The PATH Variable:

```
hacker@path~the-path-variable:~$ PATH=""
hacker@path~the-path-variable:~$ /challenge/run
Trying to remove /flag...
/challenge/run: line 4: rm: No such file or directory
The flag is still there! I might as well give it to you!
pwn.college{YcCoMN5Cw4qZrkzPog3_Tw0Fynt.QX2cDM1wyN5QTMxIzW}
hacker@path~the-path-variable:~$
```

🎉 Successfully completed The PATH Variable! 🎉

_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 :ynt.QX2cDM1wyN5QTMxIzW}



📄 sudo

2. Setting PATH:

```
hacker@path~setting-path:~$ PATH=/challenge/more_commands/  
hacker@path~setting-path:~$ /challenge/run  
Invoking 'win'....  
Congratulations! You properly set the flag and 'win' has launched!  
pwn.college{QEYgz600di9DGkHEZfMsCLuKEhz.QX1cjM1wyN5QTMxIzW}  
hacker@path~setting-path:~$
```

🎉 Successfully completed Setting PATH! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 KEhz.QX1cjM1wyN5QTMxIzW}



📄 sudo

3. Finding Commands:

```
hacker@path-finding-commands:~$ which win
/challenge/paths/24762/win
hacker@path-finding-commands:~$ cat /challenge/paths/24762/win
#!/bin/bash

/bin/fold -s <<< "Search for the flag in the same directory in which I am located!"
hacker@path-finding-commands:~$ cat /challenge/paths/24762/flag
pwn.college{YLjFQ0-C_HCUtck1Rmi-Am9i-Ln.01NzEzNxwyN5QTMxIzW}
hacker@path-finding-commands:~$
```

🎉 Successfully completed Finding Commands! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 -Ln.01NzEzNxwyN5QTMxIzW}

↻

📄 sudo

4. Adding Commands:

```
hacker@path~adding-commands:~$ echo "/bin/cat /flag" > ~/win
hacker@path~adding-commands:~$ chmod +x ~/win
hacker@path~adding-commands:~$ PATH=/home/hacker
hacker@path~adding-commands:~$ /challenge/run
Invoking 'win'....
pwn.college{AnsCpzbc0UKVarmubJRayXqDZPw.QX2cjM1wyN5QTMxIzW}
hacker@path~adding-commands:~$
```

🎉 Successfully completed Adding Commands! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩)ZPw.QX2cjM1wyN5QTMxIzW}

↻

📄 sudo

5. Hijacking Commands:

```
hacker@path-hijacking-commands:~$ echo "/bin/cat /flag" > /home/hacker/rm
hacker@path-hijacking-commands:~$ chmod +x /home/hacker/rm
hacker@path-hijacking-commands:~$ PATH=/home/hacker
hacker@path-hijacking-commands:~$ /challenge/run
Trying to remove /flag...
pwn.college{Y34QXg_w0ZjE-2R4Lmj1l96vZv6.QX3cjM1wyN5QTMxIzW}
hacker@path-hijacking-commands:~$
```

 Successfully completed Hijacking Commands!



>_ Terminal 

</> Code 

Desktop 

SSH 

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 sudo

15. Silly Shenanigans:

1. Bashrch Backdoor:

```
hacker@shenanigans~bashrc-backdoor:~$ echo "cat /flag" >> /home/zardus/.bashrc
hacker@shenanigans~bashrc-backdoor:~$ /challenge/victim
Username: zardus
pwn.college{KHdtXPaSvG2PupGy09yvsmWidpN.0VMzEzNxwyN5QTMxIzW}
zardus@shenanigans~bashrc-backdoor:~$ *****
-bash: 251722447: command not found
zardus@shenanigans~bashrc-backdoor:~$ exit

logout
hacker@shenanigans~bashrc-backdoor:~$ █
```

🎉 Successfully completed Bashrc Backdoor! 🎉

>_ Terminal [↗](#)

</> Code [↗](#)

🖥 Desktop [↗](#)

🔑 SSH ⓘ

🚩 pN.0VMzEzNxwyN5QTMxIzW}



📄 sudo

2. Sniffing Input:

```
hacker@shenanigans-sniffing-input:~$ echo 'echo "Type the flag"' > /home/hacker/
flag_checker
hacker@shenanigans-sniffing-input:~$ echo 'read FLAG' >> /home/hacker/flag_check
er
hacker@shenanigans-sniffing-input:~$ echo 'echo "You entered: $FLAG"' >> /home/h
acker/flag_checker
hacker@shenanigans-sniffing-input:~$ chmod +x /home/hacker/flag_checker
hacker@shenanigans-sniffing-input:~$ echo 'export PATH=/home/hacker:$PATH' >> /h
ome/zardus/.bashrc
hacker@shenanigans-sniffing-input:~$ /challenge/victim
Username: zardus
zardus@shenanigans-sniffing-input:~$ *****
-bash: 1955817070: command not found
zardus@shenanigans-sniffing-input:~$ flag_checker

Type the flag
*****You entered: pwn.co
lllege{sdm6mYEn_K63KxgotuJ7PL7l12s.0VNzEzNxwyN5QTMxIzW}
zardus@shenanigans-sniffing-input:~$ exit

logout
hacker@shenanigans-sniffing-input:~$
```

🎉 Successfully completed Sniffing Input! 🎉

>_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 12s.OVNzEzNxwyN5QTMxIzW}

↻

📄 sudo

3. Overshared Directories:

```
hacker@shenanigans~overshared-directories:~$ echo 'echo "Type the flag"' > /home
/hacker/flag_checker
hacker@shenanigans~overshared-directories:~$ echo 'read FLAG' >> /home/hacker/fl
ag_checker
hacker@shenanigans~overshared-directories:~$ echo 'echo "You entered: $FLAG"' >>
/home/hacker/flag_checker
hacker@shenanigans~overshared-directories:~$ chmod +x /home/hacker/flag_checker
hacker@shenanigans~overshared-directories:~$ echo 'export PATH=/home/hacker:$PAT
H' >> /home/zardus/.bashrc
bash: /home/zardus/.bashrc: Permission denied
hacker@shenanigans~overshared-directories:~$ rm /home/zardus/.bashrc
rm: remove write-protected regular file '/home/zardus/.bashrc'? y
hacker@shenanigans~overshared-directories:~$ cp /home/hacker/.bashrc /home/zardu
s/.bashrc
hacker@shenanigans~overshared-directories:~$ echo 'export PATH=/home/hacker:$PAT
H' >> /home/zardus/.bashrc
hacker@shenanigans~overshared-directories:~$ /challenge/victim
Username: zardus
Type the flag
*****Captured: 1468723600
zardus@shenanigans~overshared-directories:~$ flag_checker

Type the flag
*****You entered: pwn.co
lllege{MyPLlz7Lw6R8kePq0-23ZLaZ4l0.0FM0EzNxwyN5QTMxIzW}
zardus@shenanigans~overshared-directories:~$ exit

logout
hacker@shenanigans~overshared-directories:~$
```

 Successfully completed Overshared Directories!



 Terminal 

 Code 

 Desktop 

 SSH 

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 sudo

4. Tricky Linking:

```
hacker@shenanigans~tricky-linking:~$ rm /tmp/collab/evil-commands.txt
rm: remove write-protected regular file '/tmp/collab/evil-commands.txt'? y
hacker@shenanigans~tricky-linking:~$ ln -s /home/zardus/.bashrc /tmp/collab/evil-commands.txt
hacker@shenanigans~tricky-linking:~$ /challenge/victim
Username: zardus
zardus@shenanigans~tricky-linking:~$ *****
-bash: 2256220871: command not found
zardus@shenanigans~tricky-linking:~$ echo "cat /flag" >> /tmp/collab/evil-commands.txt

zardus@shenanigans~tricky-linking:~$ exit

logout
hacker@shenanigans~tricky-linking:~$ /challenge/victim
Username: zardus
pwn.college{YpCjDgmA9rgQm2u0V_-jIfA1oSR.0VM0EzNxwyN5QTMxIzW}
zardus@shenanigans~tricky-linking:~$ *****
-bash: 2256220871: command not found
zardus@shenanigans~tricky-linking:~$ echo "cat /flag" >> /tmp/collab/evil-commands.txt

zardus@shenanigans~tricky-linking:~$ exit

logout
hacker@shenanigans~tricky-linking:~$ █
```

🎉 Successfully completed Tricky Linking! 🎉

>_ Terminal

</> Code

Desktop

SSH

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↻

sudo

5. Sniffing Process Arguments:

```
hacker@shenanigans~sniffing-process-arguments:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.0   1056    748 ?        Ss   16:16   0:00 /sbin/docker-
root         7  0.0  0.0    7136  2836 ?        S    16:16   0:00 /run/dojo/bin
root       122  0.0  0.0   4332   2924 ?        S    16:16   0:00 su -c auto.sh
zardus     125  0.0  0.0   4752   3416 ?        Ss   16:16   0:00 /bin/bash /ru
zardus     127  0.0  0.0 235016   3512 ?        S    16:16   0:00 sleep 6h
hacker     140  0.0  0.0  37980  22836 ?        Sl   16:16   0:00 /nix/store/nz
hacker     142  0.0  0.0 232992   4980 pts/0    Ss   16:16   0:00 /run/dojo/bin
hacker     147  0.0  0.0 235228   4424 pts/0    R+   16:16   0:00 ps aux
hacker@shenanigans~sniffing-process-arguments:~$ ps -p 125 -o args=
/bin/bash /run/challenge/bin/auto.sh --user zardus --pass pw_2403632466
hacker@shenanigans~sniffing-process-arguments:~$ su zardus
Password:
zardus@shenanigans~sniffing-process-arguments:/home/hacker$ sudo cat /flag
pwn.college{E4epmETrvsrz7jH4asn02bI_44U.0FOzEzNxwyN5QTMxIzW}
zardus@shenanigans~sniffing-process-arguments:/home/hacker$
```



Successfully completed Sniffing Process
Arguments!



> Terminal

</> Code

Desktop

SSH

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sudo

6. Snooping on Configuration:

```
hacker@shenanigans~snooping-on-configurations:~$ cat /home/zardus/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
    *i*) ;;
    *) return;;
esac

# don't put duplicate lines or lines starting with space in the history.
# See bash(1) for more options
HISTCONTROL=ignoreboth

# append to the history file, don't overwrite it
shopt -s histappend

# for setting history length see HISTSIZE and HISTFILESIZE in bash(1)
HISTSIZE=1000
HISTFILESIZE=2000

# check the window size after each command and, if necessary,
# update the values of LINES and COLUMNS.
shopt -s checkwinsize

# If set, the pattern "*" used in a pathname expansion context will
# match all files and zero or more directories and subdirectories.
#shopt -s globstar

# make less more friendly for non-text input files, see lesspipe(1)
[ -x /usr/bin/lesspipe ] && eval "$(SHELL=/bin/sh lesspipe)"

# set variable identifying the chroot you work in (used in the prompt below)
if [ -z "${debian_chroot:-}" ] && [ -r /etc/debian_chroot ]; then
    debian_chroot=$(cat /etc/debian_chroot)
fi

# set a fancy prompt (non-color, unless we know we "want" color)
case "$TERM" in
    xterm-color|*-256color) color_prompt=yes;;
esac
```

```
FLAG_GETTER_API_KEY=sk-2285416452
hacker@shenanigans~snooping-on-configurations:~$ flag_getter --key 2285416452
Incorrect API key!
hacker@shenanigans~snooping-on-configurations:~$ flag_getter --key sk-2285416452
Correct API key! Do you want me to print the flag (y/n)?
y
pwn.college{Mrs8E678t0DUV_fnYDen9HX4QGx.01M0EzNxwyN5QTMxIzW}
hacker@shenanigans~snooping-on-configurations:~$
```

>_ Terminal

</> Code

Desktop

SSH

!Gx.O1M0EzNxwyN5QTMxIzW}

↺

sudo

16. Daring Destruction:

1. The Fork Bomb:

```
hacker@destruction-the-fork-bomb:~$ /challenge/check
It looks like the system can still spawn processes. We'll check again in 5 seconds...
It looks like the system can still spawn processes. We'll check again in 5 seconds...
It looks like the system can still spawn processes. We'll check again in 5 seconds...
It looks like the system can still spawn processes. We'll check again in 5 seconds...
hYou successfully saturated the process table. Here is your hard-earned flag:
pwn.college{83kRV0dXQceMAMs0x0eCP54CHPy.0VMYEzNxwyN5QTMxIzW}
hacker@destruction-the-fork-bomb:~$ h
```

Reconnecting...

🎉 Successfully completed The Fork Bomb! 🎉

> Terminal

</> Code

🖥 Desktop

🔑 SSH

📖 Description

🚩 OVMYEzNxwyN5QTMxIzW}

2. Disk-Space Doomsday:

```
hacker@destruction~disk-space-doomsday:~$ yes > big_junk_file
yes: standard output: No space left on device
hacker@destruction~disk-space-doomsday:~$ /challenge/check
Well done, you clogged the disk. Now free that space (remove the file you create
d) and run /challenge/check again to prove you cleaned up!
hacker@destruction~disk-space-doomsday:~$ rm big_junk_file
hacker@destruction~disk-space-doomsday:~$ /challenge/check
Disk space restored. Here is your flag:
pwn.college{EIkblqI1z3-aUCsxKqW4GUR-98t.0lMyEzNxwyN5QTMxIzW}
hacker@destruction~disk-space-doomsday:~$
```

🎉 Successfully completed Disk-Space Doomsday!



>_ Terminal

</> Code

Desktop

SSH

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sudo

3. rm -rf:

```
hacker@destruction-rm-rf-:~$ rm -rf /
/bin/rm: it is dangerous to operate recursively on '/'
/bin/rm: use --no-preserve-root to override this failsafe
hacker@destruction-rm-rf-:~$ rm -rf /
/bin/rm: it is dangerous to operate recursively on '/'
/bin/rm: use --no-preserve-root to override this failsafe
hacker@destruction-rm-rf-:~$ rm -rf / --no-preserve-root
/bin/rm: cannot remove '/etc/hosts': Device or resource busy
/bin/rm: cannot remove '/etc/resolv.conf': Device or resource busy
/bin/rm: cannot remove '/etc/hostname': Device or resource busy
/bin/rm: cannot remove '/usr/sbin/docker-init': Device or resource busy
/bin/rm: skipping '/sys', since it's on a different device
/bin/rm: skipping '/home/hacker', since it's on a different device
/bin/rm: skipping '/run/dojo/sys', since it's on a different device
/bin/rm: skipping '/proc', since it's on a different device
/bin/rm: skipping '/dev', since it's on a different device
/bin/rm: skipping '/nix', since it's on a different device
hacker@destruction-rm-rf-:~$
```

```
hacker@destruction-rm-rf-:~$ /challenge/check
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
Looks like you haven't wiped the system! We'll check again in 5 seconds...
YES! You wiped it, you wild hacker! The flag is yours:
pwn.college{KcNyyTw8T9kqq_2J2wt-5xieh7B.0UmZEnXwyNQ5tHxIzW}

hacker@destruction-rm-rf-:~$
```

🎉 Successfully completed `rm -rf /!` 🎉

4. Life after rm -rf:

```
hacker@destruction~life-after-rm-rf-:~$ rm -rf / --no-preserve-root
Error: refusing to remove anything under /home: /home/hacker/-
hacker@destruction~life-after-rm-rf-:~$ rm -rf --no-preserve-root /
/bin/rm: cannot remove '/etc/hosts': Device or resource busy
/bin/rm: cannot remove '/etc/resolv.conf': Device or resource busy
/bin/rm: cannot remove '/etc/hostname': Device or resource busy
/bin/rm: cannot remove '/usr/sbin/docker-init': Device or resource busy
/bin/rm: skipping '/sys', since it's on a different device
/bin/rm: skipping '/home/hacker', since it's on a different device
/bin/rm: skipping '/run/dojo/sys', since it's on a different device
/bin/rm: skipping '/proc', since it's on a different device
/bin/rm: skipping '/dev', since it's on a different device
/bin/rm: skipping '/nix', since it's on a different device
hacker@destruction~life-after-rm-rf-:~$ read -r FLAG < /flag
hacker@destruction~life-after-rm-rf-:~$ echo $FLAG
pwn.college{Q3VKkEHYP0Ic3GC87RgMw3tFBHt.01MzEzNxwyN5QTMxIzW}
hacker@destruction~life-after-rm-rf-:~$
```

🎉 Successfully completed Life after rm -rf /! 🎉

_ Terminal

</> Code

🖥 Desktop

🔑 SSH

🚩 3Ht.01MzEzNxwyN5QTMxIzW}

↻

📄 sudo

5. Finding meaning after rm -rf:

```
hacker@destruction~finding-meaning-after-rm-rf-::~$ rm -rf --no-preserve-root /
/bin/rm: cannot remove '/etc/hosts': Device or resource busy
/bin/rm: cannot remove '/etc/resolv.conf': Device or resource busy
/bin/rm: cannot remove '/etc/hostname': Device or resource busy
/bin/rm: cannot remove '/usr/sbin/docker-init': Device or resource busy
/bin/rm: skipping '/sys', since it's on a different device
/bin/rm: skipping '/home/hacker', since it's on a different device
/bin/rm: skipping '/run/dojo/sys', since it's on a different device
/bin/rm: skipping '/proc', since it's on a different device
/bin/rm: skipping '/dev', since it's on a different device
/bin/rm: skipping '/nix', since it's on a different device
hacker@destruction~finding-meaning-after-rm-rf-::~$ cd /
hacker@destruction~finding-meaning-after-rm-rf-::~$ echo *
0689d56e dev etc home nix proc run sys usr
hacker@destruction~finding-meaning-after-rm-rf-::~$ echo "$(< /0689d56e)"
pwn.college{0PQTxSwQ93zZErhXm4PanrT16G9.0FNzEzNxwyN5QTMxIzW}
hacker@destruction~finding-meaning-after-rm-rf-::~$
```



Successfully completed Finding meaning after
rm -rf /!

> Terminal

</> Code

Desktop

SSH

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sudo

CONCLUZII

Am terminat Linux Luminarium și pot spune că a fost una dintre lucrările de laborator la care am învățat cel mai mult. La început, terminalul mi se părea ceva rigid, unde greșeala unui singur caracter strică totul. Primele module, cu comenzi, argumente și căi, m-au obișnuit să-l folosesc fără să mă mai încurc în el. După ce am înțeles diferența dintre căile absolute și cele relative, am început să mă orientez în sistemul de fișiere mult mai ușor.

Modulele despre lucrul cu fișiere, documentație și globbing au fost cele în care am simțit că progrez pe bune. Comenzi ca ``cat``, ``grep``, ``find``, ``cp`` sau ``mv`` au devenit un reflex, iar ``man`` mi-a arătat că pot găsi singur răspunsul fără să caut în altă parte. Partea cu pipe-uri și redirectări mi s-a părut cea mai interesantă, pentru că am văzut cum comenzi mici se pot lega între ele și fac împreună lucruri mult mai mari. Variabilele, procesele și job-urile m-au făcut să înțeleg mai bine ce se întâmplă în spatele terminalului.

Permiisiunile și utilizatorii mi-au dat mai mult de gândit, dar au fost utile, pentru că am înțeles de ce sistemul e organizat așa. Scripturile shell, ``tmux``/``screen`` și ``PATH`` au închis cercul: am trecut de la comenzi date pe rând la lucruri automatizate. Ultimele module, cu backdoor-uri, fork bomb și ``rm -rf``, au fost și distractive, și instructive. Mi-au arătat că aceleași mecanisme care fac Linux-ul atât de flexibil pot fi folosite și greșit, de aceea trebuie să fiu atent la ce rulez și cu ce drepturi.

Acum mă simt mult mai sigur în terminal decât atunci când am început. Nu spun că le știu pe toate, dar știu unde să caut și cum să abordez o problemă, iar asta cred că îmi va fi de folos la Tehnici de programare și în orice proiect viitor pe Linux.

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2. Linux Luminarium Write-Up - <https://medium.com/@avigaildoestech/linux-luminarian-a-write-up-10e969b05213>